The Unified Hierarchical Reasoning Controller

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Abstract

The autonomous control of nonlinear systems, such as quadrotor UAVs, has traditionally required the explicit separation of slow outer-loop trajectory planning from fast inner-loop attitude stabilization. End-to-end learning approaches seek to collapse this hierarchy, but suffer from severe sample inefficiency and compounding covariate shift, which lead to divergence in dynamic environments. To address these limitations, this work develops the Unified Hierarchical Reasoning Controller (UHRC), a novel cascaded architecture which pairs a two-level recurrent transformer for high-level kinematic guidance with a classical inner-loop Proportional-Integral-Derivative (PID) attitude regulator. The UHRC is trained to generate kinematically feasible attitude intent $[\phi_{\text{ref}}, \theta_{\text{ref}}]$. This formulation inherently respects the multi-scale temporal dynamics of the plant, confining the effects of neural prediction error to the slower translational timescale, where the inner PID supplies a stabilizing correction. The resulting UHRC-PID is evaluated against a classical Simultaneous Localization and Mapping (SLAM)-augmented cascaded PID baseline in five structured test cases of increasing geometric complexity which includes open-field transit, symmetric corridor navigation, gap threading, dynamic obstacle avoidance and parametric robustness under simultaneous plant uncertainty. The hybrid achieves success rates of 20–30% in the constrained navigation scenarios and 20% under dynamic obstacle conditions, where both the pure neural policy and the classical SLAM controller record 0%. A formal cascade Input-to-State Stability (ISS) guarantee is also established for the rotational subsystem. These results establish the UHRC-PID as a robust and computationally tractable framework which resolves the divergence modes of flat end-to-end policies, while retaining the reactive spatial reasoning capabilities of learned hierarchical control.

1 Introduction

The autonomous control of complex nonlinear systems remains one of the most consequential open problems in control engineering (Iqbal et al. 2017; Z. Liu et al. 2015; Daoutidis, Megan, and W. Tang 2023). Systems such as Unmanned Aerial Vehicles (UAVs), Autonomous Ground Vehicles (UGVs) and high Degree-of-Freedom (DOF) robotic manipulators are characterized by high-dimensional state spaces, nonlinear inter-state couplings, actuator saturation and significant parametric uncertainty (Nasser Sadati, Rahmani, and

Saif 2015; Ameli and Anubi 2021; Iqbal et al. 2017; Slotine and W. Li 1991). The fundamental difficulties are that the dynamical regimes associated with long-horizon navigation and fast stabilization operate on different timescales. These dynamical regimes also differ in their sensitivity to model error and impose competing demands on the same system (Liang et al. 2018; Abdelmaksoud, Mailah, and Abdallah 2020). Designing a controller that simultaneously addresses long-horizon planning and fast reactive stabilization across these coupled dynamics is fundamentally a multiscale reasoning problem. To date, no single architecture has proven universally adequate in satisfying the competing requirements of stability, adaptability and computational tractability (Rojas, Yepes, and Garcia 2025; Goodwin, Graebe, and Salgado 2001).

Classical hierarchical control has long addressed these multiscale requirements by decomposing the control task into distinct levels. In such a formulation, a high-level supervisory layer generates strategic plans and setpoints, while a low-level regulatory layer carries out fast stabilization and trajectory tracking (Singh and Titli 1978; Nasser Sadati, Rahmani, and Saif 2015; Scattolini and Colaneri 2007; Mesarovic, Macko, and Takahara 1970; Wilson 1979). The theoretical basis for this separation traces to the multilevel systems framework of Mesarovic, Macko, and Takahara (1970) and the hierarchical decomposition methods of Singh and Titli (1978) and Siljak (2013). These works showed that large-scale optimization problems can be made tractable through their decomposition into coordinated subproblems across levels. Owing to its established stability theory, modular implementation and interpretable failure modes, hierarchical control has achieved sustained success in aerospace, process control and robotics (Sanders et al. 1998; X.-F. Wang, Jiang, and W.-H. Chen 2024; M. Liu, Tan, and Padois 2016; Bahnasawi, Al-Fuhaid, and M. Mahmoud 1990; Roberts and Becerra 2001). The quadrotor UAV directly exemplifies this paradigm. Here, the natural timescale separation between slow translational and fast rotational dynamics motivates a cascaded control structure, where an outer-loop position planner generates attitude references for an inner-loop attitude regulator (Hoffmann et al. 2007; Liang et al. 2018; S. Bouabdallah, Noth, and Siegwart 2004; Nagaty et al. 2013; Abdelhay and Zakriti 2019). Model Predictive Control (MPC) provides a systematic means of implementing the supervisory layer, permitting constraint-aware planning with formal stability guarantees at the coordination level (Scattolini and Colaneri 2007; X.-F. Wang, Jiang, and W.-H. Chen 2024; Torrente et al. 2021).

However, the fixed interface between these layers introduces a structural information bottleneck whose effects become consequential as task complexity increases (Kolchinsky, Tracey, and Wolpert 2019; S. Li et al. 2023; Abdelmaksoud, Mailah, and Abdallah 2020). The high-level planner typically communicates a heavily compressed representation, ordinarily a sequence of kinematic waypoints, to the low-level regulator. Consequently, information about obstacle proximity, actuator feasibility and local geometric constraints, which may be necessary for downstream control quality, is discarded (Zhu, Ferrari-Trecate, and Geroliminis 2024; Z. Zhou et al. 2017; W. Liu, Zheng, and Cai 2013). Conversely, the low-level regulator tracks these references without access to the global context from which they were derived, such that its response to local disturbances is globally suboptimal (Nguyen, Lowenberg, and Neild 2021; Daoutidis, W. Tang, and Allman 2019). Fixed rule-based supervisory frameworks also exhibit brittleness when operating in scenarios that were not considered during their design (N. Sadati and Ramezani 2008; Dave et al. 2023). These limitations are structural, rather than incidental. They arise from the requirement that the planner and regulator communicate through a prespecified, fixed-dimensional interface, whose information content is chosen at design time and does not adapt online

to the requirements of the task (Kolchinsky, Tracey, and Wolpert 2019; Zhang, Lu, and Gao 2015).

Learning-based approaches have emerged as a principled alternative for addressing these limitations. End-to-end Reinforcement Learning (RL) permits the joint optimization of planning and regulation without manually specifying their interface (AlMahamid and Grolinger 2021; Levine et al. 2016; Kiran et al. 2022; Shakya, Pillai, and Chakrabarty 2023). Behavioral cloning instead learns directly from expert demonstrations, providing a supervised approach that avoids the credit assignment difficulties associated with RL. Behavioral cloning has seen substantial success in autonomous driving and manipulation (Levine et al. 2016; Le Mero et al. 2022; Ly and Akhloufi 2021; Block, Foster, and Misra 2024). Flat end-to-end policies, however, are notoriously sample inefficient and sensitive to reward shaping. They do not inherently preserve the multiscale temporal structure required for long-horizon UAV navigation, and employ the same computational depth for strategic deliberation and reactive stabilization (AlMahamid and Grolinger 2021; Agyei, Sarhadi, and Polani 2025). Deep RL methods applied directly to quadrotor control have achieved high agility in simulation (Trad et al. 2024; Saviolo and Loianno 2023). Despite this, the sim-to-real gap, sample complexity and lack of formal stability guarantees remain unresolved (Dai et al. 2021; Agyei, Sarhadi, and Polani 2025). Hybrid neurosymbolic and fuzzy-integrated controllers partially address this brittleness through the combination of learned components and structured decomposition (H. H. Tang and Ahmad 2024; T. A. Mahmoud et al. 2024; Tran et al. 2021; Baheri and Alm 2025). The supervisory-regulatory interface and its dimensionality nevertheless remain manually specified. That is, the boundary between planning and regulation is fixed by the designer, rather than learned from data (Borri, Pola, and Benedetto 2019; Yu et al. 2023).

Recent developments in neural reasoning architectures offer a substantially different means of addressing this problem. Test-time scaling methods which repeat computations in latent space (Geiping et al. 2025; H. Li et al. 2025), together with recurrent architectures which adapt computational depth to task difficulty (Geiping et al. 2025; Z.-Z. Li et al. 2025), suggest that the processing depth required for a decision can be made emergent, rather than fixed. The Hierarchical Reasoning Model (HRM) of G. Wang et al. (2025) comprises two recurrent levels: a high-level module for slow, abstract planning and a low-level module for rapid, detailed computation. The modules are coupled through a shared carry state, allowing computational depth to be allocated adaptively. With as few as 1,000 training examples, and without pre-training or chain-of-thought supervision, the HRM achieves near-perfect accuracy on complex Sudoku problems and optimal pathfinding in large mazes, for which state-of-the-art Large Language Model (LLM) approaches fail entirely (G. Wang et al. 2025; J. Huang and Chang 2023; Feng et al. 2025). The hierarchical role specialization of the two modules emerges during training, rather than being prescribed by the architecture. This mirrors the hierarchy of representational dimensionality observed across cortical regions of the mammalian brain , where higher-order areas exhibit longer intrinsic timescales and higher-dimensional representations than primary sensory regions (Badre and Nee 2018; Huntenburg, Bazin, and Margulies 2018; Murray et al. 2014). Building on this, Jin et al. (2025) showed that separating planning from execution within a hierarchical framework improves multistep reasoning, further motivating the use of this principle in closed-loop control. Similarly, the hierarchical neuro-symbolic decision transformer of Baheri and Alm (2025) demonstrates that coupling a symbolic planner with a learned low-level policy through a structured interface outperforms both purely symbolic and purely neural baselines in sequential decision tasks. This result provides

further evidence that separating planning and execution offers a genuine computational advantage beyond that of either component alone.

Despite these developments, all demonstrated applications of the HRM have so far been restricted to discrete, turn-based domains. Its use as a closed-loop controller for continuous, real-time nonlinear dynamical systems remains entirely unexplored in the literature. In this setting, the control outputs must be physically continuous, observations are supplied as dense sensory streams, and temporal context must be retained over thousands of control cycles. The extension of HRMs to continuous control requires nontrivial architectural modifications. Continuous commands must replace discrete token outputs, while the supervised training signal must be obtained from a physically meaningful expert policy instead of symbolic ground truth labels. Also, localization and mapping, which are prerequisites for meaningful planning, must be supplied by an onboard sensor, as supported by Gupta and Fernando 2022; Milijas et al. 2021; Munguia, Trujillo, and Grau 2025; W. Chen et al. 2022, with its outputs incorporated into the continuous observation stream.

To address this gap, we propose and evaluate the *Unified Hierarchical Reasoning Controller* (UHRC) within a cascaded hybrid architecture for autonomous navigation of a six-degree-of-freedom quadrotor UAV. This work constitutes the first systematic application of the HRM architecture to a continuous nonlinear closed-loop control system. The UHRC adapts the two-level reasoning structure of the HRM into a recurrent transformer, which maps a 45-dimensional body-frame observation to desired attitude references $[\phi_{\text{ref}}, \theta_{\text{ref}}]$. The observation contains the kinematic state, goal displacement and a 32-ray planar Light Detection and Ranging (LiDAR) scan. Behavioral cloning, employed byLevine et al. 2016; Le Mero et al. 2022; Block, Foster, and Misra 2024 is used to train the UHRC on expert trajectories generated by an A* path planner over 500 simulated episodes (LaValle 2006; Qin et al. 2023). A classical cascaded Proportional-Integral-Derivative (PID) attitude regulator tracks the resulting references, providing formal inner-loop stability and integral disturbance rejection. In this implementation, the proposed architecture is compared with a classical hierarchical baseline comprising Simultaneous Localization and Mapping (SLAM)-based localization (Gupta and Fernando 2022; Milijas et al. 2021), A* path planning and cascaded PID control (S. Bouabdallah, Noth, and Siegwart 2004; J. Ren et al. 2016; Abdelhay and Zakriti 2019; Lopez-Sanchez and Moreno-Valenzuela 2023). The baseline's regulation layer is chosen as cascaded PID for several reasons. PID remains the dominant regulation law on embedded flight controllers (PX4, ArduPilot, and the Crazyflie's own onboard firmware), making it the more deployment relevant comparator for a system explicitly targeting resource-constrained hardware, where MPC's online optimisation is frequently infeasible at the required control rate. Further motivation is that the PID is the dominant controller in industrial practice more broadly (Åström and Hägglund 1995; Åström and Hägglund 2001). Making it the most practically representative classical baseline available. The comparison considers five structured test cases of increasing environmental complexity: open-field transit (TC01), symmetric corridor navigation (TC02), gap threading (TC03), dynamic obstacle avoidance (TC04) and parametric robustness under simultaneous plant model uncertainty (TC05).

Owing to the absence of HRM-based control methods for continuous dynamical systems, the main contributions of this work are as follows:

1. The first empirical demonstration of an HRM-based closed-loop controller for a continuous nonlinear dynamic system. This also constitutes the first use of the architecture in a hybrid cascade with a classical inner-loop regulator.

2. A hybrid UHRC-PID architecture which generates bounded, smooth attitude references through a neural outer loop, for tracking by a PD attitude regulator. This separation decouples spatial reasoning from low-level stabilization and resolves the instability caused by compounding errors in flat end-to-end behavioral cloning.

3. A formal stability analysis establishing Local Exponential Stability (LES) of the inner loop through Lyapunov synthesis, Input-to-State Stability (ISS) of the forced rotational subsystem and cascade Uniform Ultimate Boundedness (UUB) of the rotational state. The alignment condition necessary for goal convergence is also identified.

4. A rigorous comparative empirical evaluation of the performance-robustness trade-offs across all five test cases. The evaluation shows a 20% success rate under dynamic obstacle conditions, where both the pure neural policy and the classical SLAM controller fail, together with 20–30% success in constrained geometric environments.

The remainder of this paper is organized as follows. The proposed UHRC-PID architecture, observation representation, training procedure and classical baseline are detailed in Section 2. The formal cascade stability analysis is then presented in Section 2.6. Section 3 reports the quantitative results for the five test cases, followed by a discussion of the performance-robustness trade-off and its architectural origins in Section 4. Lastly, Section 5 provides the concluding remarks and directions for future work.

2 Methodology

This section presents the UHRC architecture, the classical hierarchical baseline, the expert data generation and behavioral cloning procedures, and the comparative evaluation protocol. We first describe the common 6-DOF quadrotor plant used by all controllers.

2.1 Quadrotor Dynamics Model

The benchmark system is a rigid-body quadrotor modelled with respect to the East-North-Up (ENU) inertial frame $\mathcal{I}$ and a body frame $\mathcal{B}$ fixed at the vehicle's center of mass. Its state vector is defined as

$$\mathbf{x} = \left[\mathbf{r}_I^\top,\ \mathbf{v}_I^\top,\ \mathbf{q}_{BI}^\top,\ \boldsymbol{\omega}_B^\top,\ \boldsymbol{\Omega}^\top\right]^\top \in \mathbb{R}^{17}, \tag{1}$$

where $\mathbf{r}_I \in \mathbb{R}^3$ and $\mathbf{v}_I \in \mathbb{R}^3$ denote the inertial position and velocity, respectively. The unit quaternion $\mathbf{q}_{BI} \in \mathbb{R}^4$ represents the rotation from $\mathcal{I}$ to $\mathcal{B}$ and satisfies $\|\mathbf{q}_{BI}\| = 1$, while $\boldsymbol{\omega}_B \in \mathbb{R}^3$ is the angular velocity expressed in $\mathcal{B}$ and $\boldsymbol{\Omega} \in \mathbb{R}^4$ contains the four rotor speeds. The rotor speeds are included solely for measurement.

We take the control input as the body-frame wrench

$$\mathbf{u} = \left[T,\ \tau_x,\ \tau_y,\ \tau_z\right]^\top \in \mathbb{R}^4, \tag{2}$$

which consists of the net collective thrust T along the body $+z$ axis and the roll, pitch and yaw torques τ_x, τ_y and τ_z, respectively. The formulation assumes ideal wrench actuation, bypassing the explicit rotor speed dynamics, as is appropriate for the simulation study considered here (Hoffmann et al. 2007).

Applying Newton's second law gives the inertial-frame translational dynamics as

$$m\dot{\mathbf{v}}_I = mg + R_{IB}\begin{bmatrix}0\\0\\T\end{bmatrix} + C_d^{\text{lin}}\,\mathbf{v}_I, \tag{3}$$

where m denotes the vehicle mass and g the gravitational acceleration. The matrix $R_{IB} = R_{BI}^\top$ describes the rotation from $\mathcal{B}$ to $\mathcal{I}$, obtained from $\mathbf{q}_{BI}$, and $C_d^{\text{lin}} = \text{diag}(c_x, c_y, c_z)$ is the linear aerodynamic drag matrix. The body-frame rotational dynamics are described by the Newton-Euler equations,

$$J\dot{\boldsymbol{\omega}}_B = \boldsymbol{\tau}_B - \boldsymbol{\omega}_B \times J\boldsymbol{\omega}_B - C_d^{\text{rot}}\,\boldsymbol{\omega}_B, \tag{4}$$

where $J = \text{diag}(I_{xx}, I_{yy}, I_{zz})$ is the diagonal inertia matrix, $\boldsymbol{\tau}_B = [\tau_x, \tau_y, \tau_z]^\top$, and $C_d^{\text{rot}} = \text{diag}(d_x, d_y, d_z)$ is the rotational damping matrix. The attitude evolves according to the unit quaternion differential equation,

$$\dot{\mathbf{q}}_{BI} = \tfrac{1}{2}\,(\mathbf{q}_{BI})\,\boldsymbol{\omega}_B, \tag{5}$$

where $(\mathbf{q}) \in \mathbb{R}^{4\times 3}$ denotes the standard quaternion kinematic matrix (Shastry, Kothari, and Abhishek 2018). A fourth-order Runge-Kutta scheme is used to integrate the system at a fixed timestep of $\Delta t = 0.01\,\text{s}$, with the quaternion renormalized after each step. The employed physical parameters are summarized in Table 1.

Table 1: Quadrotor physical parameters.

Parameter	Symbol	Value	Units
Mass	m	1.0	kg
Inertia (roll)	I_{xx}	0.008	$\text{kg}\,\text{m}^2$
Inertia (pitch)	I_{yy}	0.008	$\text{kg}\,\text{m}^2$
Inertia (yaw)	I_{zz}	0.014	$\text{kg}\,\text{m}^2$
Arm length	l	0.17	m
Thrust coefficient	k_T	1.9×10^{-6}	$\text{N}/(\text{rad/s})^2$
Torque coefficient	k_M	2.6×10^{-8}	$\text{N}\cdot\text{m}/(\text{rad/s})^2$
Max rotor speed	$\Omega_{\max}$	18,000	rpm
Linear drag	$\text{diag}(C_d^{\text{lin}})$	[0.10, 0.10, 0.18]	N/(m/s)
Rotational drag	$\text{diag}(C_d^{\text{rot}})$	[0.02, 0.02, 0.02]	$\text{N}\cdot\text{m}/(\text{rad/s})$

2.2 UHRC Architecture

The UHRC-PID is a two-level recurrent transformer, which adapts the HRM architecture proposed by G. Wang et al. (2025) to continuous closed-loop control. Planning and stabilization are decoupled and assigned to the UHRC and PID, respectively. At each timestep, the coupled UHRC modules produce a 2-dimensional attitude reference which expresses the strategic intent. These subgoals are supplied to the lower-level PID to generate the required torque wrench. The UHRC modules retain their memory in a rolling carry buffer, replacing the per-problem hidden state of the HRM. This permits temporal reasoning throughout the episode horizon.

The 45-dimensional body-frame observation $\mathbf{o}$ is mapped by the UHRC to the commanded attitude intent vector

$$\mathbf{u}_{\text{ref}} = [\phi_{\text{ref}}, \theta_{\text{ref}}]^\top \in \mathbb{R}^2. \tag{6}$$

The yaw reference is excluded ($\psi_{\text{ref}} = 0$), since the body-frame observations are heading-invariant and the expert commands zero yaw throughout transit. The hover thrust bias is also initialized to 0.0, as the model predicts angular deviations directly, rather than thrust forces.

The resulting attitude intent is supplied to a classical inner-loop PID regulator. This regulator computes the body wrench $\mathbf{u} = [T, \tau_x, \tau_y, \tau_z]^\top$ required to track the commanded roll and pitch, while also regulating altitude and yaw:

$$\tau_i(t) = K_{p,i} e_i(t) + K_{i,i} \int_0^t e_i(\tau) d\tau + K_{d,i} \frac{de_i(t)}{dt}, \quad i \in \{\phi, \theta\} \tag{7}$$

where $e_\phi(t) = \phi_{\text{ref}}(t) - \phi(t)$ and $e_\theta(t) = \theta_{\text{ref}}(t) - \theta(t)$.

Restricting the neural network to the outer loop removes the high-frequency stabilization task from the UHRC. The inner-loop PID then converts the generated spatial trajectories into dynamically feasible actuator commands. Thus, the architecture guarantees local asymptotic stability of the attitude dynamics while preserving the multiscale spatial reasoning capabilities of the HRM.

2.2.1 Observation Representation

The observations are expressed entirely in the body frame $\mathcal{B}$. This representation is invariant to global heading and position, reducing the domain of the mapping to be learned. That is, the same avoidance maneuver applies independently of the vehicle's absolute orientation in the world frame (Liang et al. 2018). We define the 45-dimensional observation vector as

$$\mathbf{o} = \Big[\underbrace{\mathbf{v}_B^\top}_{\times 3}, \underbrace{\boldsymbol{\omega}_B^\top}_{\times 3}, \underbrace{\mathbf{g}_B^\top}_{\times 3}, \underbrace{\Delta\mathbf{r}_B^\top}_{\times 3}, \underbrace{d_{\text{norm}}}_{\times 1}, \underbrace{\mathbf{r}_{\text{lidar}}^\top}_{\times 32}\Big]^\top \in \mathbb{R}^{45}, \tag{8}$$

where $\mathbf{v}_B = R_{BI}, v_I$ denotes the body-frame translational velocity, $\boldsymbol{\omega}_B$ the angular velocity and $\mathbf{g}_B = R_{BI}[0, 0, -g]^\top$ the gravity vector. The goal displacement $\Delta\mathbf{r}_B = R_{BI}(\mathbf{r}_{\text{goal}} - \mathbf{r}_I)$ is clipped to a maximum range of $r_{\max}^{\text{goal}} = 20\,\text{m}$, and $d_{\text{norm}} = \min(\|\Delta\mathbf{r}_B\|, 20)/20$ is the normalized goal distance. The term $\mathbf{r}_{\text{lidar}} \in [0, 1]^{32}$ denotes a 32-ray planar LiDAR scan, with field of view π rad and maximum range 5.0 m, normalized to the unit interval.

The scalar d_{norm} is included separately to retain goal-distance information. Without this term, the goal displacement vector $\Delta\mathbf{r}_B$ loses magnitude information as the drone approaches the goal, resulting in overshoot. Before being supplied to the model at runtime, the observation is standardized using precomputed dataset statistics.

2.2.2 Dual-Stream Input Encoder

Separate encoder branches process the 13 scalar kinematic features $[\mathbf{v}_B^\top, \boldsymbol{\omega}_B^\top, \mathbf{g}_B^\top, \Delta\mathbf{r}_B^\top, d_{\text{norm}}]$ and the 32-ray LiDAR scan before their outputs are fused.

The 13-dimensional input is first projected to $\mathbb{R}^{h/2}$ by a two-layer Multilayer Perceptron (MLP), using the scaled hyperbolic tangent

$$\mathbf{s} = W_2 \tanh\Big(\sqrt{H} \cdot W_1 \mathbf{o}_{\text{kin}} + \mathbf{b}_1\Big), \tag{9}$$

where $h = 128$ denotes the model's hidden dimension. The scaling factor $\sqrt{H}$ expands the effective dynamic range of the kinematic inputs into the region where tanh supplies informative gradients, thereby preventing early saturation during training.

A two-stage convolutional encoder processes the 32 normalized range returns as a 1D signal,

$$\mathbf{l} = W_{\text{proj}}\,\text{Flatten}\Big(\sigma\Big(\text{Conv}^{16}_{k=3}(\sigma(\text{Conv}^{16}_{k=5}(\mathbf{o}_{\text{lidar}})))\Big)\Big), \tag{10}$$

where σ is the Sigmoid Linear Unit (SiLU) activation and Conv^{c}_{k} denotes a 1D convolution with c output channels, kernel size k and same padding. The projection $W_{\text{proj}} \in \mathbb{R}^{(h/2)\times 512}$ maps the flattened $16 \times 32 = 512$ feature map to $\mathbb{R}^{h/2}$.

The two $h/2$-dimensional branch outputs are concatenated and linearly projected to the single embedding $\mathbf{e}_t = W_f[\mathbf{s}; \mathbf{l}] \in \mathbb{R}^h$, representing the observation at timestep t.

2.2.3 Transformer Reasoning Blocks

The H- and L-modules each comprise stacked `UHRCBlock` layers, which implement the standard pre-normalized transformer encoder block

$$\mathbf{z}' = \text{RMSNorm}\Big(\mathbf{z} + \text{MHA}(\mathbf{z})\Big), \tag{11}$$

$$\mathbf{z}'' = \text{RMSNorm}\Big(\mathbf{z}' + \text{SwiGLU}(\mathbf{z}')\Big), \tag{12}$$

where $\text{MHA}(\cdot)$ denotes non-causal multi-head self-attention with $n_h = 4$ heads and head dimension $d_h = h/n_h = 32$. The gated feed-forward network $\text{SwiGLU}(\cdot)$ is defined by

$$\text{SwiGLU}(\mathbf{z}) = W_{\downarrow}\Big(\text{SiLU}(W_g\mathbf{z}) \odot W_u\mathbf{z}\Big), \tag{13}$$

with intermediate dimension $d_{\text{ff}} = 512$. Since the carry contains a temporal history, rather than an autoregressive prediction sequence, non-causal attention is appropriate. Each position may therefore attend to every other position, including those corresponding to more recent observations.

Rotary Positional Encoding (RoPE) is applied separately to the query and key matrices of each attention head to encode the relative age of positions in the carry buffer. For sequence length T and head dimension d_h, the rotary frequencies are defined as

$$\theta_i = \frac{1}{10000^{2i/d_h}}, \quad i = 0, 1, \ldots, d_h/2 - 1, \tag{14}$$

RoPE is used in place of learned absolute positional embeddings because it generalizes to sequence positions outside those encountered during training (G. Wang et al. 2025).

2.2.4 Persistent Temporal Carry Buffer

The original HRM re-initializes the hidden states $\mathbf{z}_H$ and $\mathbf{z}_L$ for each input problem. Closed-loop control, however, requires the retention of information across successive timesteps to track velocity history, recognize developing obstacle geometries and anticipate the evolution of the trajectory. Accordingly, the UHRC represents its hidden states as persistent rolling buffers,

$$\mathbf{z}_H, \mathbf{z}_L \in \mathbb{R}^{B\times C\times h}, \tag{15}$$

where B denotes the batch size (1 at inference) and $C = 16$ the carry length. At each timestep, the observation embedding $\mathbf{e}_t$ is appended and the oldest buffer entry is removed:

$$\text{roll_carry}(\mathbf{z}, \mathbf{e}_t) = \Big[\mathbf{z}_{:,1:C,:};\ \mathbf{e}_t^{\top}\Big] \in \mathbb{R}^{B\times C\times h}, \tag{16}$$

thereby retaining a fixed-length window of temporal context without unbounded memory growth.

2.2.5 H-L Mutual Refinement and One-Step Gradient Approximation

The H- and L-modules are coupled through $H_{\text{cyc}} = 2$ outer cycles, with $L_{\text{cyc}} = 2$ inner L-module passes per cycle. At each step, concatenating the carry tokens and current observation embedding gives the full sequence $\mathbf{z}_H^{(t)} \in \mathbb{R}^{B\times 17\times h}$:

$$\mathbf{z}_H^{(t)} = \Big[\mathbf{Z}_H;\ \mathbf{e}_t^\top\Big], \quad \mathbf{z}_L^{(t)} = \Big[\mathbf{Z}_L;\ \mathbf{e}_t^\top\Big]. \tag{17}$$

The observation re-injection tensor $\mathbf{I}_{\text{obs}} \in \mathbb{R}^{B\times 17\times h}$ is constructed with zeros at the carry positions and $\mathbf{e}_t$ at position 16. Thus, the raw observation remains available as an additive bias during every L-module call (G. Wang et al. 2025).

The one-step gradient approximation used in G. Wang et al. (2025) is employed to bound the depth of the gradient graph and preserve training stability. All module calls preceding the final call in the refinement cycle are evaluated under `torch.no_grad()`, such that only a single L-pass followed by a single H-pass contributes to the gradient.

2.2.6 Subgoal Generation, Gated Injection, and Attitude Reference Output

After the final H-module update, a two-layer MLP planning head decodes the terminal hidden state $\mathbf{z}_H[:, 16, :] \in \mathbb{R}^{B\times h}$, corresponding to the most recent position in the carry sequence, into a 2-dimensional horizontal velocity subgoal in the body frame:

$$\hat{\mathbf{v}}_{\text{sub}} = W_2\,\text{SiLU}(W_1\mathbf{z}_H[:, 16, :]) \in \mathbb{R}^{B\times 2}. \tag{18}$$

This subgoal represents the H-module's strategic intent through the desired body-frame velocity vector $[v_x, v_y]$, giving a physically interpretable interface between the modules. The vertical component v_z is omitted because altitude tracking is effectively decoupled and controlled by the independent altitude regulator.

A gated additive mechanism injects the subgoal into the L-module, modulating its reactive representation without overriding it:

$$\mathbf{g} = \sigma\Big(W_g\hat{\mathbf{v}}_{\text{sub}} + \mathbf{b}_g\Big) \in (0, 1)^{B\times h}, \tag{19}$$

$$\mathbf{c} = \Big(W_p\hat{\mathbf{v}}_{\text{sub}}\Big) \odot \mathbf{g} \in \mathbb{R}^{B\times h}, \tag{20}$$

where σ denotes the element-wise sigmoid. The gate bias is initialized to $\mathbf{b}_g = +2.0$, giving $\sigma(+2) \approx 0.88$ at the beginning of training. The gate therefore remains mostly open, permitting the subgoal signal to propagate freely before convergence of the planning head. The gated context is broadcast over the carry sequence, after which the L-module carries out a final two-layer attention pass:

$$\mathbf{z}_L^{\text{final}} = \mathcal{L}\Big(\mathbf{z}_L + \mathbf{c}\cdot\mathbf{1}_{17}^\top\Big), \tag{21}$$

The attitude reference action is then decoded from the terminal position:

$$\hat{\mathbf{u}} = W_a\,\mathbf{z}_L^{\text{final}}[:, 16, :] = [\hat{\phi}_{\text{ref}},\ \hat{\theta}_{\text{ref}}]^\top \in \mathbb{R}^2. \tag{22}$$

Table 2 summarizes the complete architectural configuration.

Table 2: UHRC fixed architectural configuration.

Parameter	Symbol	Value
State dimension	d_s	45
LiDAR dimension	d_{lidar}	32
Subgoal dimension	d_{sub}	2
Hidden size	h	128
Carry length	C	16
Attention heads	n_h	4
Head dimension	d_h	32
FFN intermediate	d_{ff}	512
H-module layers	N_H	2
L-module layers	N_L	2
H-level cycles	H_{cyc}	2
L-level cycles	L_{cyc}	2
RoPE base frequency	—	10,000
Hover thrust bias	$F_z^{(0)}$	9.81 N

2.3 Expert Data Generation

The training data comprise 500 successful simulated episodes, generated by an expert policy consisting of an A* path planner, a pure-pursuit tracker and the cascaded PID attitude controller, with $\Delta t = 0.01\,\text{s}$ per step (Levine et al. 2016; Le Mero et al. 2022).

At the beginning of each episode, a 2D occupancy grid is formed with resolution $r_{\text{grid}} = 0.25\,\text{m}$ per cell. The grid spans the convex hull of the start position, goal position and all obstacle extents, together with a 2.0 m margin. Circular obstacles are inflated by $r_{\text{inf}} = 0.25\,\text{m}$ to account for the vehicle's body radius. An A* search over an 8-connected neighborhood, using the octile heuristic, guarantees the shortest collision-free path when one exists (C. Zhou, B. Huang, and Fränti 2021). Episodes without a feasible path are discarded before simulation.

Three passes of a cubic averaging filter smooth the planned path, which is then tracked by a pure-pursuit controller with lookahead $L_d = 1.0\,\text{m}$ and maximum speed $V_{\max} = 2.0\,\text{m/s}$. Within 2 m of the goal, the commanded speed is reduced linearly to 0.3 m/s. Once the vehicle enters the 0.1 m goal acceptance radius, a zero velocity command is maintained for a further 150 steps (1.5 s), demonstrating the hover termination behavior to the model.

To promote generalization of the policy beyond the default $+x$-direction transit, six categories of episodes are sampled using the distribution in Table 3.

Table 3: Episode type distribution for expert data generation.

Type	Proportion	Description
Normal	35%	Long-range $+x$ transit with 4–8 obstacles
Recovery	20%	Start beyond or beside goal; teaches re-acquisition
No obstacle	15%	Long-range transit, obstacle-free
Close	10%	Short-range (1–5 m), any direction, 0–2 obstacles
Omnidirectional	10%	Random start/goal in full arena, any heading
Tight gap	10%	Symmetric wall pair with 0.5–0.8 m half-gap

The omnidirectional episodes are necessary to eliminate the directional bias of the predominantly $+x$-transit training distribution. In their absence, the learned policy fails to navigate in the $-x$ or $\pm y$ directions.

Every 50 steps, a Gaussian velocity perturbation ($\sigma = 0.2\,\mathrm{m/s}$) and a small random yaw offset ($|\delta\psi| \leq 0.15\,\mathrm{rad}$) are applied to the current state with probability 0.3. This Dataset Aggregation (DAgger)-style augmentation, implemented by Levine et al. (2016), broadens the distribution of states encountered by the expert and prevents the compounding errors associated with pure behavioral cloning.

An episode is retained only when (i) the drone reaches the goal without collision, (ii) the goal distance decreases monotonically over coarse 100-step windows and (iii) the mean magnitude of the velocity subgoal in the final hover phase is below $0.3\,\mathrm{m/s}$. The raw A* tracker outputs are post-processed to make the expert signal learnable by the neural policy. An Exponential Moving Average (EMA) filter smooths high-frequency discontinuities in the expert commands, using smoothing factors $\alpha_{\text{act}} = 0.12$ and $\alpha_{\text{sub}} = 0.08$. A hover deadband override also demonstrates stable termination by clamping ϕ_{ref} and θ_{ref} to zero upon entry into the $0.1\,\mathrm{m}$ goal acceptance radius. The attitude commands are strictly limited to $\pm 30°$, ensuring that the demonstrations remain within the feasible limits of the low-level PID.

2.4 Behavioral Cloning Training

2.4.1 Dataset Construction

The `ControlDataset` forms sequence windows of $T_{\text{seq}} = 8$ timesteps from the stored episodes. Each window is confined to a single episode, and windows beginning within its first $T_{\text{seq}} - 1$ steps are left-padded with zeros. The 45-dimensional observations are z-scored using the dataset-wide mean and standard deviation, while the attitude references remain in their original physical units. Training and validation subsets are separated by episode, with 10% reserved for validation.

2.4.2 Loss Function

The total loss is defined as a weighted sum of the action and subgoal tracking errors, scaled by a spatial proximity multiplier:

$$\mathcal{L} = \gamma_{\text{spat}} \Big(\mathcal{L}_{\text{act}} + \alpha_s \mathcal{L}_{\text{sub}} + \alpha_t \mathcal{L}_{\text{trk}} \Big), \tag{23}$$

where $\alpha_s = 0.1$ and $\alpha_t = 0.0$ are the selected loss hyperparameters. The spatial factor γ_{spat} increases the penalty by $10\times$ when the vehicle is within a normalized distance threshold of the goal.

The action imitation loss is formulated as a weighted Mean Squared Error (MSE) of the attitude vector,

$$\mathcal{L}_{\text{act}} = \Big\langle \Big(\hat{\mathbf{u}}_{\text{ref}} - \mathbf{u}^*_{\text{ref}} \Big)^\top \operatorname{diag}(\mathbf{w}_a) \Big(\hat{\mathbf{u}}_{\text{ref}} - \mathbf{u}^*_{\text{ref}} \Big) \Big\rangle, \tag{24}$$

where the normalized action weights are initialized equally to $\mathbf{w}_a = [10.0, 10.0]$. The intermediate horizontal velocity intent is supervised through the corresponding subgoal loss:

$$\mathcal{L}_{\text{sub}} = \Big\langle \Big(\hat{\mathbf{v}}_{\text{sub}} - \mathbf{v}^*_{\text{sub}} \Big)^\top \operatorname{diag}(\mathbf{w}_s) \Big(\hat{\mathbf{v}}_{\text{sub}} - \mathbf{v}^*_{\text{sub}} \Big) \Big\rangle, \tag{25}$$

where $\mathbf{w}_s = [10.0, 10.0]$ gives the corresponding subgoal weights.

2.4.3 Optimizer and Schedule

2.4.4 Training Configuration and Episodic Streaming

The UHRC is trained through episodic streaming, with AdamW optimization over 20 epochs. The learning rate follows a linear warmup over 5 epochs, followed by cosine annealing decay.

A base weight decay of 1×10^{-2} penalizes large weights to prevent overfitting.

Table 4: Behavioral cloning training configuration.

Parameter	Symbol	Value
Sequence length	T_{seq}	8
Batch size	B	512
Peak learning rate	η	2×10^{-4}
Weight decay	λ	1×10^{-2}
Gradient clip	—	1.0
Warmup epochs	—	5
Total epochs	—	20
Validation split	—	10% (by episode)
Subgoal loss weight	α_s	0.1
Action weights	$\mathbf{w}_a$	[10.0, 10.0]
Subgoal weights	$\mathbf{w}_s$	[10.0, 10.0]

2.5 Comparative Evaluation Protocol

The controllers are compared over five structured test cases (TC01–TC05) of increasing environmental complexity. Both receive ideal state feedback from the same simulation state to ensure a fair comparison. Each case comprises $N = 30$ randomized Monte Carlo trials, with performance metrics averaged over successful and failed trials.

TC01 – Open Transit. An obstacle-free, straight-line goal-seeking task establishes the baseline trajectory tracking performance.

TC02 – Symmetric Corridor. Six obstacles are placed symmetrically about the nominal flight path to evaluate multi-obstacle avoidance and path quality under spatial constraints.

TC03 – Gap Navigation. A 3 m gap formed by two obstacles lies directly in the nominal flight path, permitting evaluation of control precision and gap-threading capability.

TC04 – Dynamic Obstacle Avoidance. Obstacles are withheld from the controller's sensor data until $t \geq 3.0$ s, when their revelation requires reactive replanning during flight. Identical sensor data are supplied to both controllers at every timestep.

TC05 – Parametric Robustness. The obstacle arrangement of TC02 is retained while the plant parameters are perturbed: mass by ±20%, diagonal inertia by ±30%, drag coefficients by ±50% and thrust coefficient k_T by ±15%. This test characterizes each controller's robustness margin under distributional shift.

Performance is evaluated using the Root Mean Square Error (RMSE) of the Euclidean position error, Integral Absolute Error (IAE), Integral Time-weighted Absolute Error (ITAE), Path Length Ratio (PLR = actual path length / straight-line distance) and

episode success rate. An episode is successful when the vehicle reaches within 0.5 m of the goal without collision.

2.6 Stability Analysis

The neural policy supplies attitude references to a fixed-gain, torque-saturated inner loop. We establish regional input-to-state stability for the ideal rotational plant under initial-state, reference-increment, and disturbance constraints, and a separate active-saturation result for pure-axis motion. Position convergence and collision avoidance require additional arguments.

2.6.1 Standing Assumptions

Let $\eta = (\phi, \theta, \psi)^\top$ and $\omega_B = (p, q, r)^\top$ denote Euler angles and body angular rates, respectively. In particular, q here is a body rate, not a quaternion. The rotational plant is

$$\dot{\eta} = \mathcal{T}(\phi, \theta)\omega_B, \qquad J\dot{\omega}_B = \tau - D\omega_B - \omega_B \times (J\omega_B) + w(t), \tag{26}$$

where

$$\mathcal{T}(\phi, \theta) = \begin{bmatrix} 1 & \sin\phi\tan\theta & \cos\phi\tan\theta \\ 0 & \cos\phi & -\sin\phi \\ 0 & \sin\phi/\cos\theta & \cos\phi/\cos\theta \end{bmatrix}. \tag{27}$$

Here $K_{p,i}, K_{d,i}$ denote the components of the gain vectors and J_i, d_i the diagonal entries of J, D.

At $t_k = kh$, a reference $\eta_{r,k} = (\phi_{r,k}, \theta_{r,k}, 0)^\top$ i s selected and held until t_{k+1}. On a fixed smooth angle-error branch, the controller computes

$$e_{i,k} = \eta_{r,i,k} - \eta_i(t_k), \qquad v_{i,k} = K_{p,i}e_{i,k} - K_{d,i}\omega_i(t_k), \qquad \tau_{i,k} = \mathrm{sat}_{[-\bar{\tau}_i, \bar{\tau}_i]}(v_{i,k}). \tag{28}$$

The torque $\tau_{i,k}$ is held on $[t_k, t_{k+1})$. This is measured-rate damping; no derivative-of-reference feedforward is included. The attitude integral state is zero and all attitude integral gains remain zero.

Assumption 1: The operating domain has $|\phi|, |\theta| \le \pi/6$, a fixed smooth angle-error branch, and torque limits $(0.6, 0.6, 0.35)$ Nm. The certificate uses ideal applied body torques. Torque commands are clipped componentwise to these limits and held on each sampling interval, h.

Assumption 2: The physical rotational model is

$$J\dot{\omega}_B = \tau - D\omega_B - \omega_B \times (J\omega_B) + w(t), \qquad \dot{\eta} = \mathcal{T}(\eta)\omega_B,$$

where D is the rotational damping matrix. The external torque w is measurable and satisfies

$$\operatorname*{ess\,sup}_{t \ge 0} \|w(t)\|_2 \le W.$$

The undisturbed model is recovered by taking $w = 0$.

Assumption 3: Applied references are selected at $t_k = kh$ and held between updates, with $\eta_{r,k} = (\phi_{\mathrm{ref},k}, \theta_{\mathrm{ref},k}, 0)^\top$. They satisfy

$$|\phi_{\mathrm{ref},k}|, |\theta_{\mathrm{ref},k}| \le \bar{r} \le \pi/6, \qquad \|\eta_{r,k+1} - \eta_{r,k}\|_2 \le R.$$

Assumption 4: Actual thrust satisfies $0 \le T(t) \le T_{\max}$.

2.6.2 Inner-Loop Dynamics

Use the scaled tracking state

$$\chi = (e_\phi, p/\sigma_\phi, e_\theta, q/\sigma_\theta, e_\psi, r/\sigma_\psi)^\top, \qquad \sigma = (37.2, 37.2, 8.31). \tag{29}$$

Let $\chi_k = \chi(t_k^+)$ denote the state after selecting $\eta_{r,k}$. Writing $T_{r,k} = \mathcal{T}(\phi_{r,k}, \theta_{r,k})$ and $\delta\eta_{r,k} = \eta_{r,k+1} - \eta_{r,k}$.

Lemma 1 (Regional representation)**.** *If $|v_{i,k}| \le 2\bar{\tau}_i$, there exists $\lambda_{i,k} \in [1/2, 1]$ such that $\tau_{i,k} = \lambda_{i,k} v_{i,k}$.*

Proof. For $|v_{i,k}| \le \bar{\tau}_i$, choose $\lambda_{i,k} = 1$. Otherwise choose $\lambda_{i,k} = \bar{\tau}_i/|v_{i,k}|$. The assumed requested-torque bound places this ratio in $[1/2, 1]$. The multiplier is fixed throughout the hold interval. □

Define $C \in \mathbb{R}^{3\times 6}$ by $C_{i,2i-1} = K_{p,i}$ and $C_{i,2i} = -K_{d,i}\sigma_i$, with other entries zero. Then $v_k = C\chi_k$ and $\Lambda_k = \mathrm{diag}(\lambda_{1,k}, \lambda_{2,k}, \lambda_{3,k})$. For $0 \le t \le h$, let

$$b_i(t) = e^{-d_i t/J_i}, \qquad f_i(t) = \frac{1 - b_i(t)}{d_i/J_i}, \qquad g_{1,i}(t) = \frac{t - f_i(t)}{d_i}, \qquad g_{2,i}(t) = \frac{f_i(t)}{J_i}. \tag{30}$$

For each interval, $F_{t,k}, G_{t,k}$ are defined by their nonzero entries:

$$\begin{aligned} (F_{t,k})_{2i-1,2j-1} &= \delta_{ij}, & (F_{t,k})_{2i-1,2j} &= -(T_{r,k})_{ij}\sigma_j f_j(t), \\ (F_{t,k})_{2i,2j} &= \delta_{ij} b_i(t), & (G_{t,k})_{2i-1,j} &= -(T_{r,k})_{ij} g_{1,j}(t), \\ (G_{t,k})_{2i,j} &= \delta_{ij}\frac{g_{2,i}(t)}{\sigma_i}. \end{aligned} \tag{31}$$

Set $L_{t,k} = F_{t,k} + G_{t,k}\Lambda_k C$ and $N_k = L_{h,k}$. The fixed injection maps are

$$\begin{aligned} \mathcal{E}u &= (u_1, 0, u_2, 0, u_3, 0)^\top, \\ \mathcal{B}u &= (0, u_1/(J_1\sigma_1), 0, u_2/(J_2\sigma_2), 0, u_3/(J_3\sigma_3))^\top. \end{aligned} \tag{32}$$

and define the exact nonlinear remainder as

$$\mathcal{R}(\chi, \eta_{r,k}) = -\mathcal{E}[(\mathcal{T}(\eta) - T_{r,k})\omega_B] + \mathcal{B}[-\omega_B \times (J\omega_B)]. \tag{33}$$

For $0 \le t < h$, variation of constants gives

$$\chi(t_k + t) = L_{t,k}\chi_k + \int_0^t F_{t-v,k}\left[\mathcal{R}(\chi(t_k + v), \eta_{r,k}) + \mathcal{B}w(t_k + v)\right] dv. \tag{34}$$

At the endpoint, Taking the left limit and then applying the reference jump:

$$\begin{aligned} \chi_{k+1}^- &= N_k\chi_k + \int_0^h F_{h-v,k}\left[\mathcal{R}(\chi(t_k + v), \eta_{r,k}) + \mathcal{B}w(t_k + v)\right] dv, \\ \chi_{k+1} &= \chi_{k+1}^- + \mathcal{E}\delta\eta_{r,k}. \end{aligned} \tag{35}$$

Thus reference increments enter directly, without differentiating a discontinuous held reference.

2.6.3 Common Quadratic Certificate and Finite Verification

Take $P = \mathrm{diag}(P_\phi, P_\theta, P_\psi)$, with all displayed decimals interpreted exactly, where

$$P_\phi = \begin{bmatrix} 1.9473235487 & -2.1836575679 \\ -2.1836575679 & 6.0335073972 \end{bmatrix},$$
$$P_\theta = \begin{bmatrix} 8.7782076491 & -2.5300374186 \\ -2.5300374186 & 13.9909634365 \end{bmatrix},$$
$$P_\psi = \begin{bmatrix} 1.2679581274 & -1.9194171927 \\ -1.9194171927 & 14.7490226386 \end{bmatrix}. \tag{36}$$

Define $z = \|\chi\|_P = \sqrt{\chi^\top P \chi}$ and the induced matrix norm $\|A\|_P = \|P^{1/2} A P^{-1/2}\|_2$.

Proposition 1 (Numerical map certificate)**.** *The matrix in* (36) *satisfies* $0.999 I_6 \prec P \prec 15.018 I_6$. *For all* $|\phi_{r,k}|, |\theta_{r,k}| \le \pi/6$, $\lambda_{i,k} \in [1/2, 1]$, *and* $t \in [0, h]$,

$$\|N_k\|_P < \rho = 0.9951, \qquad \|L_{t,k}\|_P < c_L = 1.048, \qquad \|F_{t,k}\|_P < c_F = 1.152. \tag{37}$$

Proof. The numerical matrix inequalities are verified by directed interval arithmetic, using the exact displayed decimal coefficients. Partition each reference angle over $[-0.524, 0.524]$ at $-0.262, 0, 0.262$. The results cover $[-\pi/6, \pi/6]^2$. Establishing

$$\rho^2 P - N_k^\top P N_k \succ 0. \tag{38}$$

Interval $LDL^\top$ factorization gives a strictly positive lower bound for each pivot; the smallest verified sampled pivot lower bound exceeds 0.01445. The triangle inequality for the common induced norm extends (38) to each full box.

For intersample coverage, $1.149^2 P - F_{h,k}^\top P F_{h,k} \succ 0$. Since $f_i''(t) = -(d_i/J_i) e^{-d_i t/J_i}$, the chord interpolation defect obeys

$$\epsilon_i(t) = f_i(t) - \frac{t}{h} f_i(h), \qquad 0 \le \epsilon_i(t) \le \frac{d_i h^2}{8 J_i}. \tag{39}$$

There are rank-one matrices $B_{i,k}$ such that

$$L_{t,k} = \left(1 - \frac{t}{h}\right) I_6 + \frac{t}{h} N_k + \sum_{i=1}^{3} \epsilon_i(t) B_{i,k}. \tag{40}$$

In grouped coordinates $(e^\top, (\omega_B/\sigma)^\top)^\top$, their factors are

$$B_{i,k} = \mathbf{u}_{i,k} \mathbf{v}_{i,k}^\top, \quad \mathbf{u}_{i,k} = \begin{bmatrix} T_{r,k}[:, i]/d_i \\ \mathbf{e}_i/(J_i \sigma_i) \end{bmatrix}, \quad \mathbf{v}_{i,k}^\top = \begin{bmatrix} \lambda_{i,k} K_{p,i} \mathbf{e}_i^\top & -\big(\lambda_{i,k} K_{d,i} + d_i\big) \sigma_i \mathbf{e}_i^\top \end{bmatrix}. \tag{41}$$

Here $\mathbf{e}_i$ is a coordinate basis vector. The metric is permuted consistently when using these grouped coordinates. Verified bounds on the primal and dual factor norms are respectively $(58.64, 133.65, 92.00)$ and $(8.84, 5.83, 2.114)$. Because $\|uv^\top\|_P = \sqrt{u^\top P u}\sqrt{v^\top P^{-1} v}$, their summed contribution in (40) is less than 0.047262. Consequently $\|L_{t,k}\|_P < 1.047262 < 1.048$. The analogous open-flow interpolation defect is below 0.001955, giving $\|F_{t,k}\|_P < 1.149 + 0.001955 < 1.152$. These estimates cover the entire hold interval. □

Remark 1. *The same P applies to reference and saturation variation within the stated family.*

2.6.4 Nonlinear Remainder and Physical-State Bounds

For the axis block $\chi_i = (e_i, \omega_i/\sigma_i)^\top$, put $z_i = \sqrt{\chi_i^\top P_i \chi_i}$, so that $z^2 = z_\phi^2 + z_\theta^2 + z_\psi^2$. The inequality

$$|a^\top x| \le \sqrt{a^\top P^{-1} a}\, \|x\|_P \tag{42}$$

and direct block inversion give the following conservative coefficients:

Bound	Roll	Pitch	Yaw
$\lvert e_i\rvert \le a_i z_i$	$a_\phi = 0.930$	$a_\theta = 0.347$	$a_\psi = 0.992$
$\lvert\omega_i\rvert \le \beta_i z_i$	$\beta_\phi = 19.65$	$\beta_\theta = 10.22$	$\beta_\psi = 2.415$
$\lvert C_i\chi\rvert \le t_i z$	$t_\phi = 8.68$	$t_\theta = 5.66$	$t_\psi = 2.066$

The injection-map bounds are

$$\|\mathcal{E}\delta\eta_r\|_P \le c_E \|\delta\eta_r\|_2, \quad c_E = 2.963 \quad (\delta\eta_{r,3} = 0), \qquad \|\mathcal{B}w\|_P \le c_B \|w\|_2, \quad c_B = 32.37. \tag{43}$$

Lemma 2 (Remainder bound)**.** *If actual and reference roll and pitch lie in* $[-\pi/6, \pi/6]$*, then*

$$\|\mathcal{R}(\chi, \eta_r)\|_P \le C_R \|\chi\|_P^2, \qquad C_R = 25. \tag{44}$$

Proof. Let $P_E = \mathrm{diag}(P_{11}, P_{33}, P_{55})$ and split $\mathcal{R} = \mathcal{R}_{\mathrm{kin}} + \mathcal{R}_{\mathrm{gyro}}$ according to (33). The first column of either angle derivative of $\mathcal{T}$ is zero. Bounding the spectral norm by the Frobenius norm gives

$$\begin{aligned} \left\|P_E^{1/2}(\partial_\phi \mathcal{T})[:, 2{:}3]\right\|_2^2 &\le P_{11}/3 + P_{33} + 4P_{55}/3 < 3.335^2, \\ \left\|P_E^{1/2}(\partial_\theta \mathcal{T})[:, 2{:}3]\right\|_2^2 &\le 16P_{11}/9 + 4P_{55}/9 < 2.007^2. \end{aligned} \tag{45}$$

These estimates use $\tan^2\theta \le 1/3$ and $\sec^2\theta \le 4/3$. The line segment joining actual and reference angles remains in the angle box. The mean-value formula therefore yields

$$\begin{aligned} \|\mathcal{R}_{\mathrm{kin}}\|_P &\le (3.335|e_\phi| + 2.007|e_\theta|)\sqrt{q^2 + r^2} \\ &\le \left[\frac{3.335(0.930)(10.22)}{2} + 2.007(0.347)(10.22)\right] z^2 = 22.96642488 z^2. \end{aligned} \tag{46}$$

Here $z_\phi\sqrt{z_\theta^2 + z_\psi^2} \le z^2/2$, while $z_\theta\sqrt{z_\theta^2 + z_\psi^2} \le z^2$.

Axisymmetric inertia gives

$$J^{-1}[-\omega_B \times (J\omega_B)] = \gamma_J(-qr, pr, 0)^\top, \qquad \gamma_J = \frac{0.01428 - 0.0082}{0.0082}. \tag{47}$$

Since $\sqrt{P_{22}}/37.2 < 0.06604$ and $\sqrt{P_{44}}/37.2 < 0.10056$, the rate bounds and $z_\psi\sqrt{z_\phi^2 + z_\theta^2} \le z^2/2$ imply

$$\|\mathcal{R}_{\mathrm{gyro}}\|_P \le \frac{\gamma_J(2.415)}{2} \max\{0.06604(10.22), 0.10056(19.65)\} z^2. \tag{48}$$

The sum of the two coefficients is less than $24.735575 < 25$. The triangle inequality proves (44). □

Remark 2. *The remainder bound is conditional on the stated angle domain. Its coefficient depends on the metric and scaling.*

2.6.5 Neural Reference Map: Boundedness and Regularity

Let $\mathbf{o}_k \in \mathbb{R}^{n_o}$ denote the observation, with n_o equal to the deployed model's configured observation dimension, and let $m_k \in \mathcal{Z} = \mathbb{R}^{2\times C_{\mathrm{mem}}\times H}$ denote its stored carry. Here C_{mem} is the carry length and H is the hidden dimension. The feedback matrix C retains its definition above. The applied roll/pitch reference map is

$$f_{\mathrm{UHRC}}(\mathbf{o}_k, m_k) = \mathrm{sat}_{[-\pi/6,\pi/6]^2}\Big(f_{\mathrm{raw}}(\mathbf{o}_k, m_k)\Big), \qquad \eta_{r,k} = \begin{bmatrix} f_{\mathrm{UHRC}}(\mathbf{o}_k, m_k) \\ 0 \end{bmatrix}. \tag{49}$$

Lemma 3 (Output bound)**.** *For every finite evaluation,* $f_{\mathrm{UHRC}}(\mathbf{o}, m) \in [-\pi/6, \pi/6]^2$.

Proof. Componentwise saturation bounds the applied reference. □

Lemma 4 (Local Lipschitz regularity)**.** *For fixed finite weights, frozen observation normalization with positive scales, positive RMSNorm regularization, and a fixed finite number of reasoning updates,* f_{UHRC} *is Lipschitz on compact convex observation–carry sets.*

Proof. The affine maps, attention, smooth gates, and regularized normalization in the finite inference computation are smooth, with bounded derivatives on each such set and on its intermediate images. Saturation is nonexpansive in the Euclidean norm, composition gives a finite local Lipschitz constant. For the unweighted normalization

$$\mathrm{RMSNorm}(x) = \frac{x}{\sqrt{\|x\|_2^2/H + \epsilon}}, \qquad \epsilon > 0,$$

the Jacobian has spectral norm at most $1/\sqrt{\epsilon}$. A quantitative constant for the complete policy must also account for observation normalization, attention, residual additions, products, and the repeated reasoning steps. □

In the implemented architecture, each newly stored carry token is an output of unweighted RMSNorm and therefore has Euclidean norm at most $\sqrt{H}$ in exact arithmetic. Finite initialized tokens are either retained or replaced by such tokens, so the stored carry remains in a bounded finite-dimensional set.

The interconnection theorem additionally requires the applied reference sequence to satisfy the stated amplitude and increment constraints.

2.6.6 Coupled Attitude Tracking: Regional ISS and Invariance

Admissible references and convex tracking-state regions. Choose $0 < s < S$, $0 \le \bar{r} < \pi/6$, and $R, W \ge 0$. Let the reference and disturbance family satisfy

$$\begin{aligned} |\phi_{r,k}|, |\theta_{r,k}| &\le \bar{r}, & \psi_{r,k} &= 0, \\ \|\delta\eta_{r,k}\|_2 &\le R, & \operatorname*{ess\,sup}_{t_k \le t < t_{k+1}} \|w(t)\|_2 &\le W. \end{aligned} \tag{50}$$

Define

$$\Omega_s = \{\chi : \chi^\top P \chi \le s^2\}, \qquad \Omega_S = \{\chi : \chi^\top P \chi \le S^2\}. \tag{51}$$

Both are convex ellipsoids because $P \succ 0$. For example, for $\chi_1, \chi_2 \in \Omega_s$ and $0 \le \alpha \le 1$,

$$\|\alpha\chi_1 + (1-\alpha)\chi_2\|_P \le \alpha \|\chi_1\|_P + (1-\alpha)\|\chi_2\|_P \le s.$$

Invariance additionally requires the following inequalities.

Put

$$d_h = c_F h C_R S, \qquad c_w = c_F h c_B, \qquad a_* = \rho + \frac{c_L d_h}{1 - d_h}, \qquad b_w = \frac{c_w}{1 - d_h}. \tag{52}$$

Define the admissible parameter family $\mathcal{A}$ by

$$0 < s < S, \quad 0 \le \bar{r} < \pi/6, \quad R, W \ge 0, \quad d_h < 1, \quad a_* < 1, \tag{53}$$

$$\bar{r} + 0.930S \le \pi/6, \qquad \frac{c_L s + c_w W}{1 - d_h} < S, \tag{54}$$

$$c_E R + b_w W \le (1 - a_*)s, \tag{55}$$

$$8.68s \le 1.2, \qquad 5.66s \le 1.2, \qquad 2.066s \le 0.7. \tag{56}$$

The final line constrains requested torques at ticks to the regional sector. The actual torques remain clipped at $(0.6, 0.6, 0.35)$ Nm. No convexity claim is made for the nonlinear parameter-feasibility set $\mathcal{A}$.

Coupled regional ISS theorem.

Theorem 1 (Sampled invariance, intersample containment, and regional ISS)**.** *Consider* (26)–(28) *under the standing assumptions, with a tuple* $(\bar{r}, s, S, R, W) \in \mathcal{A}$ *and inputs satisfying* (50)*. If* $\chi_0 \in \Omega_s$*, then* $\chi_k \in \Omega_s$ *for every* $k \ge 0$*, and* $\|\chi(t)\|_P < S$ *between updates. Actual roll and pitch remain in* $[-\pi/6, \pi/6]$*. Moreover, with* $z_k = \|\chi_k\|_P$*,*

$$z_k \le a_*^k z_0 + \frac{1 - a_*^k}{1 - a_*}(c_E R + b_w W). \tag{57}$$

Proof. Suppose $z_k \le s$. The component bounds and (56) place the held torque in the sector of Lemma 1. Consider a hypothetical first exit from the S-tube during this interval. Up to that time, the actual angle bounds follow from $|\phi|, |\theta| \le \bar{r} + 0.930S \le \pi/6$, so Lemma 2 applies. For the largest weighted radius Y up to that time, (34) gives

$$Y \le c_L z_k + c_F h C_R S Y + c_F h c_B W = c_L z_k + d_h Y + c_w W. \tag{58}$$

Since $d_h < 1$,

$$Y \le \frac{c_L z_k + c_w W}{1 - d_h} \le \frac{c_L s + c_w W}{1 - d_h} < S,$$

contradicting the first exit. Existence extends across the interval because the state remains in a compact region away from the Euler singularity and the disturbance is essentially bounded.

At the next tick, (35) and the bounds on N_k and $\mathcal{E}$ yield

$$\begin{aligned} z_{k+1} &\le \rho z_k + d_h Y + c_w W + c_E \left\|\delta\eta_{r,k}\right\|_2 \\ &\le a_* z_k + b_w W + c_E R \le a_* s + (1 - a_*)s = s. \end{aligned} \tag{59}$$

Induction establishes sampled invariance and intersample containment for all intervals. Iterating (59) and summing a geometric series proves (57). Finally, $|\phi| \le |\phi_r| + |e_\phi|$ and $|\theta| \le |\theta_r| + |e_\theta|$ establish actual-angle containment. □

Corollary 1 (Ultimate bound and zero-input convergence)**.** *Under Theorem 1,*

$$\limsup_{k\to\infty} \|\chi_k\|_P \le \frac{c_E R + b_w W}{1 - a_*}. \tag{60}$$

Let $p_- = \lambda_{\min}(P)$ *and* $p_+ = \lambda_{\max}(P)$*. Then*

$$\|\chi_k\|_2 \le \sqrt{\frac{p_+}{p_-}} a_*^k \|\chi_0\|_2 + \frac{1 - a_*^k}{(1 - a_*)\sqrt{p_-}}(c_E R + b_w W). \tag{61}$$

For a constant admissible reference and $w = 0$*, the tracking state converges exponentially at samples and between samples. Persistent nonzero admissible inputs generally imply an ultimate neighborhood rather than convergence exactly to the origin.*

Proof. Since $0 < a_* < 1$, its powers tend to zero, proving (60). Norm equivalence gives (61). When $R = W = 0$, $z_k \le a_*^k z_0$ and $\|\chi(t_k + t)\|_P \le c_L a_*^k z_0/(1 - d_h)$ for $0 \le t < h$. Writing $\alpha_* = -\log(a_*)/h > 0$ gives a continuous-time exponential bound with a finite prefactor, because $a_*^{\lfloor t/h \rfloor} \le a_*^{-1} e^{-\alpha_* t}$. □

Parametric budgets. For each $(\bar{r}, s, S, W)$ satisfying the remaining conditions, the allowed reference increment is

$$0 \le R \le R_{\max}(s, S, W) := \frac{(1 - a_*)s - b_w W}{c_E}, \qquad b_w W \le (1 - a_*)s. \tag{62}$$

The intersample condition additionally requires

$$0 \le W < \frac{(1 - d_h)S - c_L s}{c_w}. \tag{63}$$

Thus region size, disturbance allowance, and reference increments cannot be chosen independently. The torque-sector conditions constrain the controller's requested torque. w is a separate external torque in the plant equation and is accounted for in the tube and input budgets.

The component bounds give the physical estimates

$$\begin{aligned} |\phi(t)| &\le \bar{r} + 0.930S, \quad |\theta(t)| \le \bar{r} + 0.347S, \\ \|\omega_B(t)\|_2 &\le 19.65S, \qquad |\tau_{i,k}| \le \min\{\bar{\tau}_i, t_i s\}, \end{aligned} \tag{64}$$

where $(t_\phi, t_\theta, t_\psi) = (8.68, 5.66, 2.066)$. The rate estimate follows from $\|\omega_B\|_2^2 \le \max_i \beta_i^2 \sum_i z_i^2$.

Numerical specialization. One admissible tuple is

$$\bar{r} = 29.5^\circ, \qquad s = 0.0075, \qquad S = 0.0083, \qquad R = 6 \times 10^{-6} \text{ rad/update}, \qquad W = 0.$$

It gives $d_h = 0.0023904$, $a_* < 0.997611142$, and $Y < 0.007878834 < S$. Using that conservative rounded bound on a_*,

$$c_E R = 0.000017778 < 0.000017916435 \le (1 - a_*)s.$$

The actual-angle bound is $|\phi|, |\theta| < 29.94227^\circ$, and requested torques are bounded by $(0.06510, 0.04245, 0.015495)$ Nm. The increment-equivalent rate is $R/h \approx 0.0343775^\circ/\text{s}$.

Remark 3 (Scope of the coupled certificate)**.** *For the constants above, $a_* < 1$ itself requires*

$$S < \frac{1-\rho}{c_F h C_R (c_L + 1 - \rho)} < 0.01616. \tag{65}$$

Because $s < S$, every feasible coupled tube obtained from these inequalities has requested torques below $(0.14027, 0.09147, 0.03339)$ Nm, strictly below the saturation limits. Thus this coupled certificate, although formulated for the saturated law, only establishes regions in which saturation is inactive. Its small reference-increment allowance does not certify ordinary rapid coupled navigation. Failure of the inequalities outside this family does not prove physical instability.

2.6.7 Pure-Axis Tracking

Consider either exact pure-roll motion with $(\theta, \psi, q, r) = (0, 0, 0, 0)$, or exact pure-pitch motion with $(\phi, \psi, p, r) = (0, 0, 0, 0)$. Set the other two attitude references to zero and take $w = 0$. Each such subspace is invariant for the ideal plant. Write ϑ for the selected angle and ν for its body rate. Then $\dot{\vartheta} = \nu$ and $J_a \dot{\nu} = \tau - d_a \nu$, with $J_a = 0.0082$, $d_a = 0.022$.

Let $x = (e, \nu/37.2)^\top$, $e = r_k - \vartheta$, and $c = (11.3425, -0.49581(37.2))$. Using the scalar functions in (30) at h, put

$$F_h^a = \begin{bmatrix} 1 & -37.2f \\ 0 & b \end{bmatrix}, \qquad G_h^a = \begin{bmatrix} -g_1 \\ g_2/37.2 \end{bmatrix}, \qquad E_a = \begin{bmatrix} 1 \\ 0 \end{bmatrix}. \tag{66}$$

The exact sampled system is

$$x_{k+1} = F_h^a x_k + G_h^a \,\mathrm{sat}_{[-0.6,0.6]}(c x_k) + E_a (r_{k+1} - r_k). \tag{67}$$

For

$$P_a = \begin{bmatrix} 1 & -0.8 \\ -0.8 & 1.7 \end{bmatrix}, \qquad \det(P_a) = 1.06, \tag{68}$$

interval verification establishes

$$(F_h^a + \lambda G_h^a c)^\top P_a (F_h^a + \lambda G_h^a c) \prec 0.855^2 P_a, \qquad \lambda \in \{1/2, 1\}. \tag{69}$$

The minimum verified pivot lower bound exceeds 0.0023027. Convexity extends the norm inequality to every $\lambda \in [1/2, 1]$. The accompanying bounds are

$$\begin{aligned} \sqrt{c P_a^{-1} c^\top} &< 14.543, & |e| &\le 1.267 \,\|x\|_{P_a}, \\ |\nu| &\le 36.132 \,\|x\|_{P_a}, & \|L_t^a\|_{P_a} &< 1.022 \quad (0 \le t \le h). \end{aligned} \tag{70}$$

Here $L_t^a = F_t^a + \lambda G_t^a c$ for the multiplier selected at the tick.

Theorem 2 (Saturated tracking on an exact pure-axis invariant subspace)**.** *Assume either of the pure-axis initial-state and reference restrictions stated in Section 2.6.7, with zero external torque, fixed parameters and ideal held body torque. Suppose*

$$|r_k| \le \pi/9, \qquad |r_{k+1} - r_k| \le 0.01, \qquad \|x_0\|_{P_a} \le 0.08,$$

where the increment is in radians per update. Then $\|x_k\|_{P_a} \le 0.08$ at every sample, $\|x(t)\|_{P_a} < 0.08176$ between samples, and $|\vartheta(t)| < 25.94^\circ < 30^\circ$ for all time.

Proof. At a tick inside the ellipsoid, $|cx_k| < 14.543(0.08) = 1.16344 < 1.2$, so the regional sector applies. Since $\|E_a\|_{P_a} = 1$,

$$\|x_{k+1}\|_{P_a} \le 0.855\,\|x_k\|_{P_a} + 0.01 \le 0.0784 < 0.08.$$

Within the hold interval, $\|x(t)\|_{P_a} < 1.022(0.08) = 0.08176$. Thus

$$|\vartheta(t)| \le \frac{\pi}{9} + 1.267(1.022)(0.08) < \frac{25.936\pi}{180} < \frac{\pi}{6}.$$

Iteration also gives

$$\limsup_{k\to\infty} \|x_k\|_{P_a} \le \frac{0.01}{1 - 0.855} \approx 0.068966.$$

□

Remark 4 (Active saturation)**.** *The initial state*

$$x_0 = 0.08\frac{P_a^{-1}c^\top}{\sqrt{cP_a^{-1}c^\top}}$$

lies on the ellipsoid boundary and requests approximately 1.16339 *Nm, which is clipped to* 0.6 *Nm. Active saturation is therefore included in Theorem 2. Its reference increment corresponds to* $0.01/h = 1$ *rad/s, or approximately* $57.2958°/s$*. This pure-axis result cannot be applied independently to simultaneous roll and pitch motion, which reintroduces the omitted coupling.*

2.6.8 Scope of the UHRC Interconnection

Let the applied neural reference be $\eta_{r,k} = \mathrm{sat}(f_{\mathrm{raw}}(o_k, m_k))$, where m_k denotes the stored neural memory and the yaw reference is set as specified above. Amplitude clipping bounds the requested angles for finite outputs. It does not establish actual-angle containment, the increment bound $\|\delta\eta_{r,k}\|_2 \le R$, or the required initial tracking state. The theorems apply to UHRC only when the applied sequence is shown or enforced to satisfy the relevant admissible reference family. The quantity R/h is an increment-equivalent rate; a zero-order-held reference has jumps and does not need a bounded classical derivative. Similarly, bounded attitude tracking does not by itself establish bounded position error, navigation convergence, or obstacle avoidance.

Remark 5 (Numerical verification and implementation consistency)**.** *The matrix and remainder constants above belong to the fixed displayed configuration. Parameter changes require fresh verification of every dependent matrix inequality, residual bound, input budget, and physical containment condition. In particular, headroom must be evaluated as* $\pi/6 - \bar{r}$*.*

2.6.9 Velocity Bounds and Conditional Goal-Region Attractiveness

Proposition 2 (Velocity UUB)**.** *Assume additionally that* $m > 0$ *and the horizontal dynamics satisfy*

$$m\dot{\mathbf{v}}_I^{\mathrm{h}} = F_h - C_h\mathbf{v}_I^{\mathrm{h}}, \qquad \|F_h(t)\|_2 \le T_{\max}, \qquad C_h = C_h^\top \succeq c_{\min} I_2, \quad c_{\min} > 0.$$

For example, the force bound follows from an actual thrust bound $0 \le T(t) \le T_{\max}$ *when no additional horizontal force is present. Then*

$$\|\mathbf{v}_I^{\mathrm{h}}(t)\| \le e^{-c_{\min}t/m}\|\mathbf{v}_I^{\mathrm{h}}(0)\| + \frac{T_{\max}}{c_{\min}}(1 - e^{-c_{\min}t/m}).$$

Thus $\limsup_{t\to\infty} \|\mathbf{v}_I^{\mathrm{h}}(t)\| \le T_{\max}/c_{\min}$*, but attitude stability and bounded velocity alone do not guarantee bounded position error.*

Proof. The horizontal force satisfies $\|F_h\| \le T_{\max}$, giving

$$D^+\|\mathbf{v}_I^{\mathrm{h}}\| \le T_{\max}/m - (c_{\min}/m)\|\mathbf{v}_I^{\mathrm{h}}\|.$$

Here D^+ is the upper right Dini derivative. Scalar comparison gives the estimate. A constant horizontal force can produce nonzero terminal velocity and unbounded position drift. □

Theorem 3 (Conditional goal-region attractiveness)**.** *Let* $V_{\mathrm{nav}} = \frac{1}{2}\|\mathbf{r}_I^{\mathrm{h}} - \mathbf{r}_{\mathrm{goal}}^{\mathrm{h}}\|^2$*. For* $r_0 > 0$*, assume the goal region* $\mathcal{G} = \{\|\mathbf{r}_I^{\mathrm{h}} - \mathbf{r}_{\mathrm{goal}}^{\mathrm{h}}\| \le r_0\}$ *is forward invariant under the complete closed-loop dynamics, whose trajectories are assumed to exist for all subsequent times. If there exists a class-*$\mathcal{K}$ *function* α_3*, continuous and strictly increasing with* $\alpha_3(0) = 0$*, such that*

$$\dot{V}_{\mathrm{nav}} = (\mathbf{r}_I^{\mathrm{h}} - \mathbf{r}_{\mathrm{goal}}^{\mathrm{h}})^\top \mathbf{v}_I^{\mathrm{h}} \le -\alpha_3(V_{\mathrm{nav}}) \tag{71}$$

outside $\mathcal{G}$ *for all* $t \ge T$*, the trajectory enters* $\mathcal{G}$ *in finite time and remains there.*

Proof. Outside $\mathcal{G}$, $\dot{V}_{\mathrm{nav}} \le -\alpha_3(r_0^2/2) < 0$. Failure to enter would force V_{nav} below $r_0^2/2$ in finite time, a contradiction. Invariance ensures persistence. □

Remark 6. *Equation* (71) *is a sufficient policy-dependent condition, not established by the attitude certificate or guaranteed by behavioural cloning. An obstacle detour may temporarily violate it.*

3 Results

This section reports the quantitative and qualitative performance of the UHRC-PID and classical hierarchical baseline over the five test cases (TC01–TC05). The metrics are computed from $N = 30$ Monte Carlo trials per scenario. RMSE and IAE are evaluated over the complete duration of each episode, which depends on the time required to reach the goal, and include failed and divergent trials. This is done so that the aggregate statistics characterize operational reliability, rather than selective best-case performance. The PLR is the ratio of the actual path length travelled to the straight-line distance between the start and goal. Table **??** summarizes the performance of both controllers across all scenarios.

3.1 Classical Hierarchical Baseline

The classical baseline comprises four stages which implement the full supervisory-regulatory decomposition described in the literature (Sanders et al. 1998; Scattolini and Colaneri 2007). It serves as a representative instance of the class of hierarchical architectures that the UHRC is intended to replace.

An Inertial Navigation System Extended Kalman Filter (INSEKF) combines Global Positioning System (GPS) position estimates with Attitude and Heading Reference System (AHRS) orientation estimates at 10 Hz, yielding the pose estimate $[\hat{\mathbf{r}}_I, \hat{\mathbf{q}}_{BI}]$. LiDAR odometry is obtained through Iterative Closest Point (ICP) registration of 2D point cloud keyframes. The keyframes are accumulated in a pose graph, which is optimized using the Levenberg-Marquardt algorithm at fixed intervals of 500 nodes (Gupta and Fernando 2022; Milijas et al. 2021).

The A* algorithm then computes a collision-free path on a 2D occupancy grid, formed by projecting the LiDAR returns onto the ground plane with a cell resolution of 2.0 m. The grid covers a height band of ±2.0 m about the drone's estimated flight level. Path feasibility is guaranteed when a collision-free grid path exists (C. Zhou, B. Huang, and Fränti 2021); otherwise, the previously valid path is retained.

The planned waypoint sequence is converted into position setpoints at 10 Hz by a pure pursuit tracker with fixed lookahead distance $L_d = 0.8$ m (Y. Huang et al. 2020).

Controller	Scenario	Success (%)	RMSE (m)	IAE (m s)	ITAE (m s^2)	PLR
Classical	TC01	100.0	0.938	43.426	1120.854	2.090
	TC02	100.0	0.6306	11.25	129.54	1.158
	TC03	100.0	0.631	9.313	83.941	1.158
	TC04	0.0	—	—	—	—
	TC05	100.0	0.707	13.346	144.978	1.227
UHRC-PID	TC01	30.0	5.307	99.282	933.290	2.10
	TC02	25.0	7.193	73.128	513.464	1.09
	TC03	20.0	9.250	95.919	802.358	1.47
	TC04	20.0	5.425	73.193	542.541	1.866
	TC05	33.3	5.696	124.82	1665.458	2.55

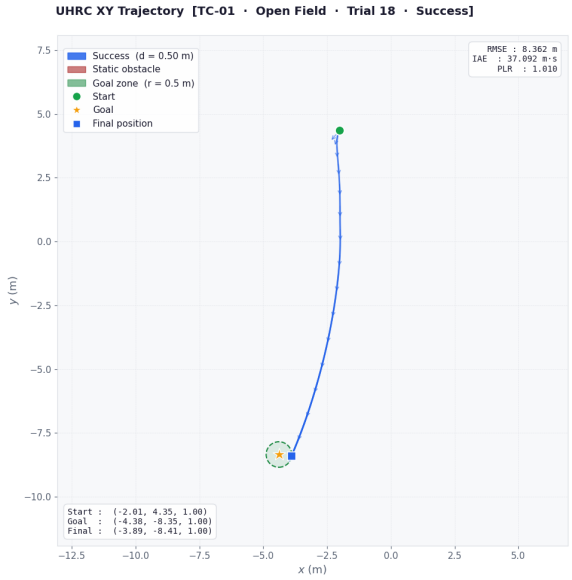
3.2 TC01: Open-Field Transit

TC01 evaluates baseline goal-seeking performance in an obstacle-free environment. The classical controller reached the goal in all 30 trials, achieving 100% success, with a mean RMSE of 0.938 m and an IAE of 43.426 m s. The residual tracking error is attributable to cumulative drift in the ICP odometry subsystem and the fixed-interval pose graph optimization. During square-trajectory validation, the INSEKF position estimator exhibited residual errors of 0.376 m along x and 0.265 m along y, confirming the direct propagation of localization noise into tracking error. The PLR of 2.090 results from lookahead-induced lateral oscillations of the pure-pursuit tracker about the nominal straight-line path.

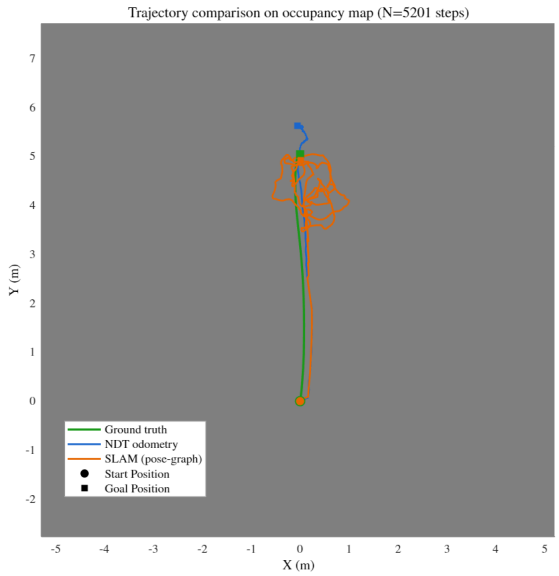
The UHRC-PID achieved 30% success (9 of 30 trials), with a mean RMSE of 5.307 m and a PLR of 2.09. In the two successful trials, the initial goal distance was below approximately 7 m. At this range, the policy converged to a stable goal-seeking attractor before the carry state drifted outside the training distribution. For more distant goals, however, errors in the generated torque predictions accumulate over successive timesteps, driving the vehicle into states not previously encountered by the model. Figures 1a and 1b show the qualitative trajectories of both controllers.

3.3 TC02: Symmetric Corridor Navigation

In TC02, six obstacles are arranged symmetrically about the nominal flight corridor, requiring a collision-free path to be planned and executed within a geometrically constrained environment. The classical controller achieved 100% success and a PLR of 1.158,



(a) UHRC-PID trajectory (Success: 30%, RMSE: 5.307 m)



(b) Classical trajectory (Success: 100%, RMSE: 0.938 m)

Figure 1: TC01 open-field transit trajectories. The UHRC diverges from the goal in 90% of trials, while the classical controller converges in all 30 trials.

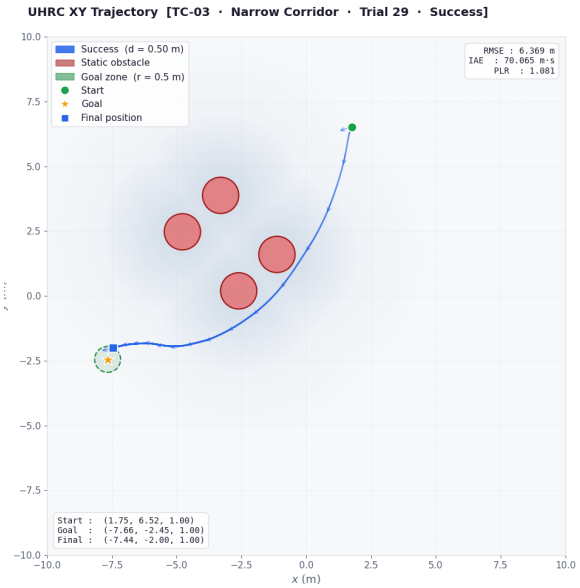
indicating that A* found a valid corridor path in every trial. The modest elevation of the PLR above 1.0 arises because the 2.0 m occupancy grid resolution prevents precise path placement in narrow passages, necessitating small detours. The RMSE of 0.6306 m and IAE of 11.25 m s confirm accurate tracking of the planned path, as shown in Figure 2b.

The UHRC achieved 25% success (7 of 30 trials), with a mean RMSE of 7.193 m, an IAE of 73.128 m s and a PLR of 1.09. Despite this low success rate, Figure 2a shows a smooth and directed approach trajectory around the corridor. This indicates that the UHRC does not directly represent the corridor as a navigable passage. The trajectory instead suggests that the trained policy mapped the LiDAR readings to large attitude references, resulting in the observed suboptimal flight path.

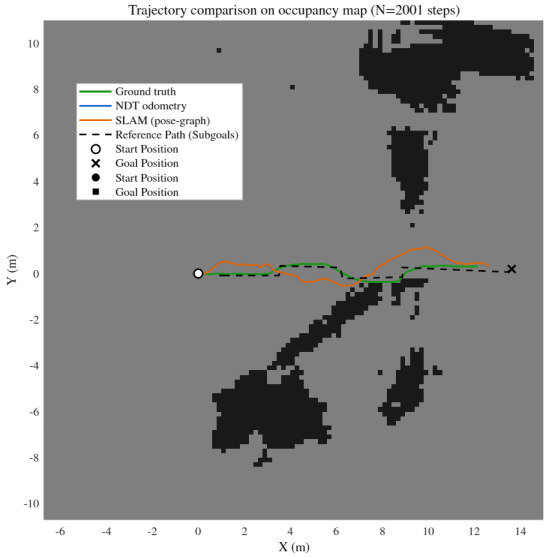
3.4 TC03: Gap Navigation

TC03 evaluates gap identification and the control precision needed for collision-free traversal by placing two obstacles with a 3 m separation directly in the nominal flight path. The classical controller achieved 100% success, with an RMSE of 0.631 m and a PLR of 1.158. A* identified the gap and planned a path through it in every trial. Near-miss events were nevertheless observed during several gap transits. These are attributable to waypoint discretization at the 2.0 m grid resolution and replanning latency at 10 Hz, which together prevent precise path placement relative to the obstacle surfaces.

The UHRC achieved 20% success, with a mean RMSE of 9.25 m and a PLR of 1.47. Figure 3a shows a smooth approach trajectory directed accurately toward the gap, indicating that the UHRC correctly represents the gap as a navigable passage.



(a) UHRC-PID trajectory (Success: 25%, RMSE: 7.193 m)



(b) Classical trajectory (Success: 100%, PLR: 1.158)

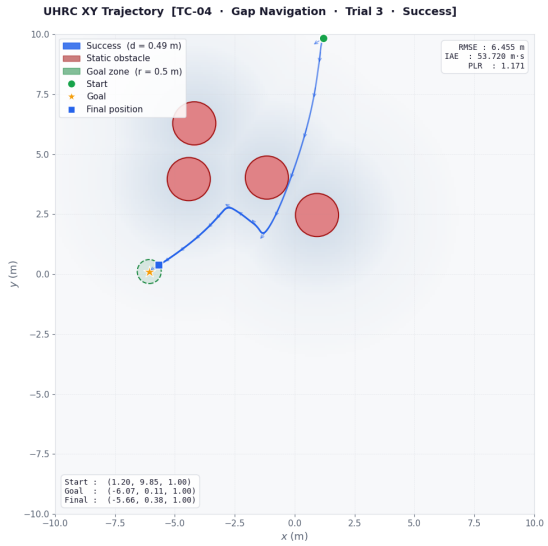
Figure 2: TC02 symmetric corridor trajectories.

3.5 TC04: Dynamic Obstacle Avoidance

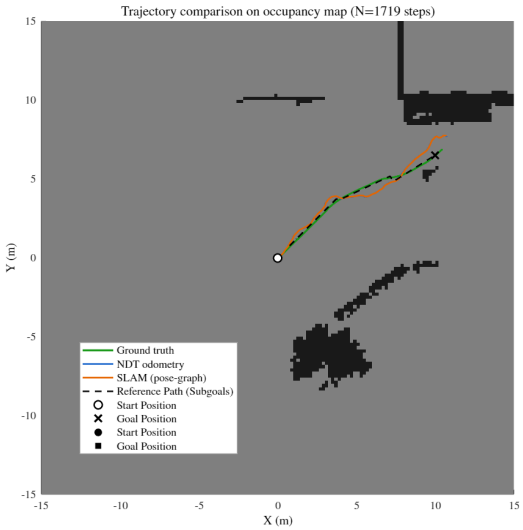
TC04 provides the most significant finding of the evaluation. Obstacles are withheld from both sensor streams until $t \geq 3.0\,\text{s}$, when they are revealed during flight. The classical controller failed in all 30 trials, giving a 0% success rate, as shown in Figure 4.

The failure of the classical controller follows from its architecture. Obstacle detection requires projection of the LiDAR scan into the occupancy grid, computation of a revised A* waypoint sequence and propagation of that sequence to the pure-pursuit tracker. A delay therefore occurs between obstacle revelation and the availability of an updated reference trajectory. During this interval, the low-level PID continues to track the previous trajectory, causing collision before replanning can redirect the vehicle, as confirmed by Figure 4b. The delay is inherent to the fixed supervisory-regulatory interface and cannot be removed by tuning the existing architecture.

Failures of the UHRC-PID in this test case are attributable to the policy instability that dominates all test cases. The drone diverges from its nominal path before reaching the obstacle, instead of colliding with it. Small erratic attitude predictions appear immediately after carry initialization, consistent with the accumulation of out-of-distribution states before obstacle revelation at $t = 3.0\,\text{s}$. Notably, however, in trials where a coherent flight path was maintained until $t = 3.0\,\text{s}$, as shown in Figure 4a, the 32-ray LiDAR scan immediately registered the newly revealed obstacle, and the observation embedding was updated within the same H-L reasoning cycle. The resulting replanning latency is bounded by a single 10 ms inference call, compared with the 100 ms pipeline cycle. This direct response constitutes the principal structural advantage of unified reasoning over explicit hierarchical decomposition in dynamic scenarios.



(a) UHRC trajectory (Success: 20%, RMSE: 9.25 m)



(b) Classical trajectory (Success: 100%, RMSE: 0.631 m)

Figure 3: TC03 gap navigation trajectories. Both controllers direct the vehicle toward the 3 m gap.

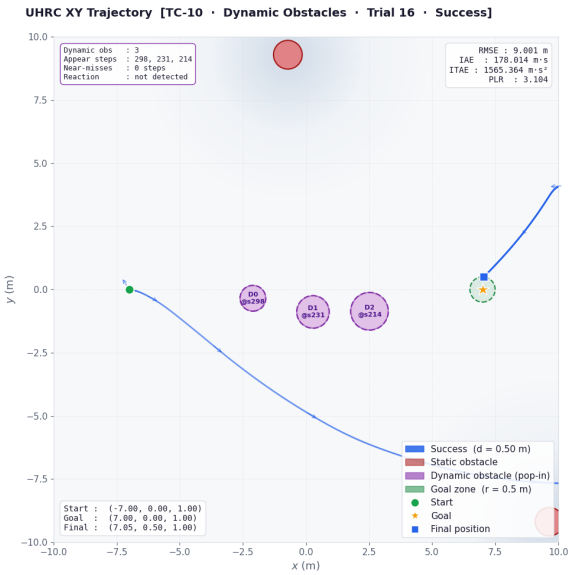
3.6 TC05: Parametric Robustness

TC05 retains the corridor arrangement of TC02 while simultaneously perturbing the plant parameters (mass ±20%, diagonal inertia ±30%, drag coefficients ±50%). The contrasting responses of the controllers characterize the performance-robustness boundary between learned and classical architectures.

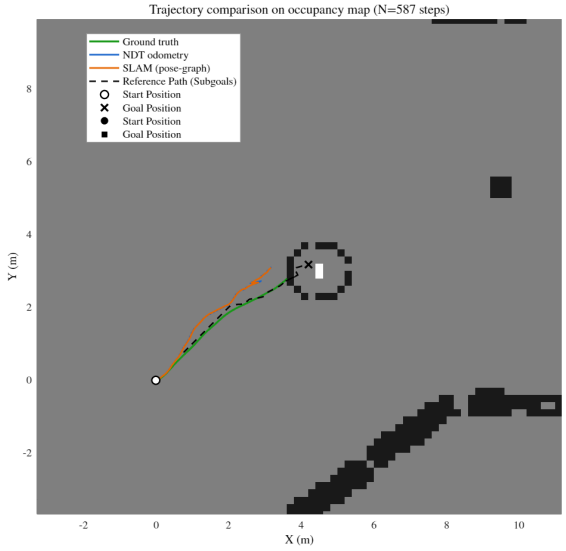
As shown in Figure 5, the classical controller achieved 100% success over all 30 perturbed trials, with a mean RMSE of 0.707 m, an IAE of 13.346 m s and a PLR of 1.227. The metrics remain comparable to the nominal TC01 and TC03 results, demonstrating bounded performance degradation despite significant parametric variation. The thrust output under perturbed conditions (2814.94 Nm) was elevated relative to the nominal TC01 baseline (5136.5 Nm), which is consistent with the integral term actively compensating for the reduced effective hover thrust.

Under TC05 conditions, the UHRC achieved 33.3% success (10 of 30 trials), with a mean RMSE of 58.675 m and an ITAE of 73,011.560 m s^2. Figure 6 compares the UHRC trajectories under nominal and perturbed dynamics. The nominal case (Figure 6a) shows a coherent straight-line transit, whereas 20% model uncertainty (Figure 7) causes the trajectory to deviate from the intended path. The error metrics are less informative in this comparison because the trials use different initial positions and goals. Nevertheless, the qualitative trajectory comparison shows that the perturbed parameters produce the suboptimal path.

Under perturbed conditions, the torque magnitude in Figure 7 exhibits large oscillations about the nominal hover torque. This behavior is expected, since the PID performs low-level stabilization, but the UHRC torque predictions are not robust to parametric uncertainty. When small prediction errors occur in the attitude references, the PID loop compensates for them through corrective attitude action.



(a) UHRC-PID trajectory (Success: 20%, RMSE: 5.425 m). The vehicle diverges from the nominal path before the obstacle is revealed at $t = 3.0$ s.



(b) Classical controller trajectory (Success: 0%). The vehicle maintains the prior planned trajectory after obstacle revelation due to replanning pipeline latency, resulting in collision.

Figure 4: TC04 dynamic obstacle avoidance trajectories ($N = 30$ trials).

4 Discussion

The results from TC01–TC05 reveal a consistent picture of the two control architectures. Failures of the classical controller are architecturally deterministic, owing to the replanning delay imposed by the supervisory-regulatory interface, which cannot be reduced without redesigning the pipeline. The UHRC-PID, meanwhile, exhibits superior reactive spatial reasoning in dynamic environments, relying on the inner-loop PID for rigid-body stabilization while the outer-loop neural network handles kinematic guidance. Their differing modes of success and failure therefore identify complementary limits of the two designs.

4.1 Policy Stability and Guidance Efficacy

The cascaded implementation provides a proof of principle that the UHRC can achieve highly effective goal-directed navigation under nominal conditions. The inner-loop PID performs high-frequency attitude regulation, removing the need for the outer loop to compensate for fast rotational dynamics. By extension, this mitigates the accumulation of distributional errors (Le Mero et al. 2022; Block, Foster, and Misra 2024).

The classical inner loop supplies a continuous restoring force against transient deviations through integral correction. In a representative open-field transit (TC01), the resulting hybrid achieved a PLR of 1.010, an IAE of 37.092 m·s and an RMSE of 8.362 m. These metrics confirm the functional correctness of the learned spatial reasoning when unbounded attitude divergence is prevented. Thus, the spatial planning component of the UHRC has the potential for stable long-horizon closed-loop operation when decoupled from low-level action generation.

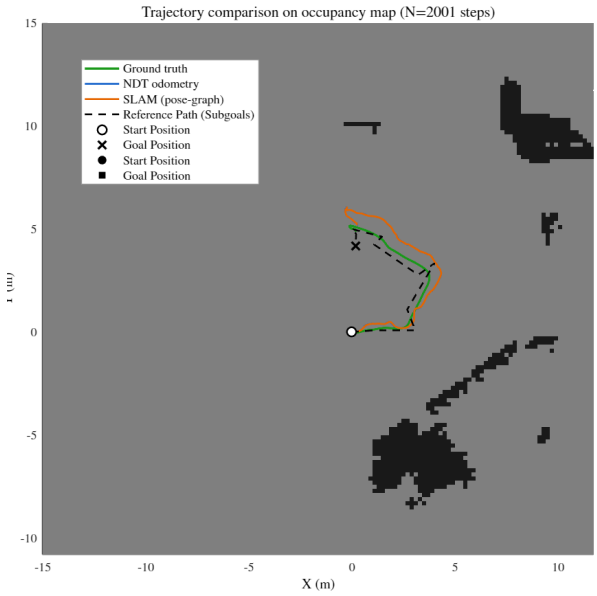


Figure 5: Classical controller trajectory under parametric uncertainty (TC05, Success: 100%, RMSE: 0.707 m).

4.2 Overcoming the Information Bottleneck: TC02 and TC03

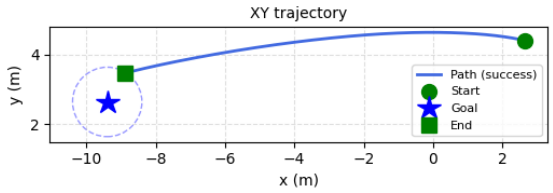
The theoretical motivation for the UHRC is to resolve the information bottleneck at the classical supervisory-regulatory interface. The A* planner supplies only a waypoint sequence to the low-level PID, discarding all contextual information associated with obstacle proximity and local geometry (Kolchinsky, Tracey, and Wolpert 2019; Abdelmaksoud, Mailah, and Abdallah 2020). The TC02 and TC03 results provide strong empirical support for this hypothesis. Near obstacle surfaces, the classical controller oscillates laterally about the planned path. This behavior is attributable to the pure-pursuit tracker receiving only positional waypoints, without any representation of clearance from neighboring obstacles.

The UHRC-PID trajectories exhibit qualitatively smoother clearance geometry in the constrained scenarios. This agrees with the L-module having direct access to LiDAR returns through the unified observation vector when computing attitude intent. During symmetric corridor transit (TC02), for example, the UHRC-PID achieved a PLR of 1.081, although the observed flight path was suboptimal relative to that of the classical controller. During gap navigation (TC03), the corresponding PLR was 1.171 and the RMSE was 6.455 m. Because basic policy stability is maintained by the inner loop, the UHRC successfully demonstrates its architectural advantage in constrained geometric environments.

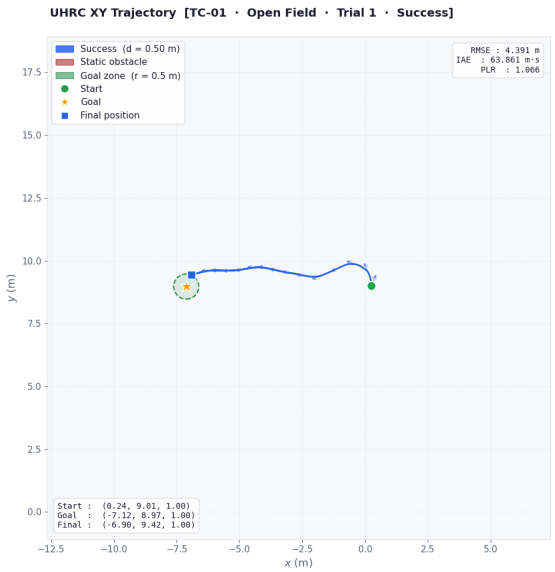
4.3 Architectural Highlights in Dynamic Environments: TC04

TC04 provides the most architecturally significant result of the evaluation, demonstrating where the learned hierarchy explicitly outperforms classical explicit decomposition. Obstacles are withheld from both controllers' sensor streams until $t \geq 3.0\,\mathrm{s}$ and are then revealed in the drone's active flight path.

The classical controller must first project the LiDAR scan into the occupancy grid, recompute the A* waypoint sequence and transmit the revised path to the pure-pursuit



(a) UHRC under nominal dynamics (100% model certainty).



(b) UHRC under perturbed dynamics (80% model certainty).

Figure 6: TC05 UHRC trajectory comparison between nominal and perturbed plant dynamics.

tracker. The resulting delay causes collision before replanning can redirect the vehicle (S. Li et al. 2023; Zhu, Ferrari-Trecate, and Geroliminis 2024). This delay is inherent to the fixed supervisory-regulatory interface and cannot be addressed through tuning.

The UHRC-PID, however, successfully navigated the dynamic environment. A representative successful trial achieved a PLR of 3.104, an IAE of 178.014 m·s and an RMSE of 9.001 m. The LiDAR observation was updated during the same 10 ms inference cycle in which the obstacle was revealed, with the new geometry immediately represented in the observation embedding of the next H-L reasoning cycle. Bounding the replanning latency by a single inference call gives unified reasoning its principal structural advantage over explicit hierarchical decomposition in dynamic scenarios.

4.4 Distributional Fragility vs Integral Robustness: TC05

TC05 most directly characterizes the fundamental trade-off between the architectures under simultaneous perturbations of mass (±20%), inertia (±30%), drag (±50%) and thrust coefficient (±15%).

The classical controller's success demonstrates that its cascaded integral feedback is a principled and highly effective mechanism for parametric robustness. The PID integrators treat changes in the effective plant dynamics as persistent disturbances, compensating for them continuously without requiring knowledge of their magnitude.

The UHRC-PID achieves similar performance, although some distributional fragility remains. Training through behavioral cloning on a single nominal dynamic model calibrates the policy's implicit representation of spatial kinematics exclusively to the nominal parameter set (Le Mero et al. 2022). Under perturbed conditions, the control inputs exhibit the expected increase in thrust, maintaining approximately 11.8 N compared with the nominal hover thrust of 9.81 N. Despite the attitude bounds imposed during training, prediction errors in the attitude references still produce high-frequency torque oscillations.

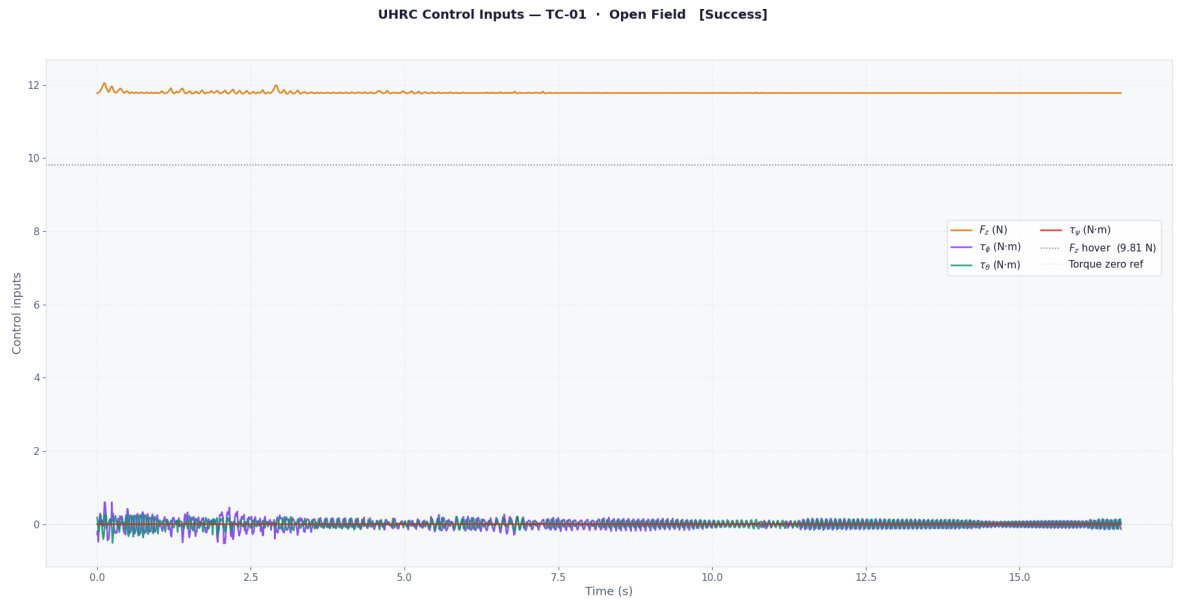


Figure 7: UHRC total torque magnitude under 80% model certainty (TC05).

The PID compensates for these errors, but the overall trajectory remains suboptimal.

Despite the uncertainties, a successful perturbed transit achieved a PLR of 1.066, an IAE of 63.861 m·s and an RMSE of 4.391 m. This shows that grounding the execution in classical PID logic anchors the degraded kinematic guidance of the HRM, providing a significantly wider stability margin even as the outer-loop predictions deteriorate.

5 Conclusion

In this work, we developed the Unified Hierarchical Reasoning Controller (UHRC), constituting the first systematic application of the Hierarchical Reasoning Model (HRM) architecture to a continuous nonlinear closed-loop control system. The model was ultimately formulated as a cascaded UHRC-PID hybrid, pairing a recurrent transformer outer loop, which generates attitude intent, with a classical inner loop for high-frequency rigid-body stabilization. Comparison with a classical SLAM-augmented cascaded PID baseline over five structured test cases established the principal findings of the study. Firstly, the HRM architecture can learn a highly functional continuous control policy when timescale separation is respected. Secondly, the cascaded formulation successfully addresses the compounding errors which constitute the main limitation to reliable deployment of monolithic end-to-end networks. Thirdly, single-step inference allows the unified architecture to bypass structural replanning delays and outperform the classical baseline in dynamic environments (TC04). Finally, although the hybrid improves general robustness, classical integral feedback remains superior in rejecting severe, unmodelled parametric uncertainty.

Future work will primarily consider domain randomization during training, which directly addresses the remaining distributional fragility without modifying the architecture. Replacement of the PID regulators with an ideal controller will also be investigated to improve the accuracy of the learned model. RL fine-tuning from the behaviorally cloned initialization would further permit the discovery of corrective strategies absent from the offline expert. Beyond these extensions, the UHRC framework provides a systematic

basis for investigating whether unified hierarchical reasoning can serve as the definitive outer-loop guidance architecture for complex autonomous systems.

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